Prepared for submission to Phys. Rev. D.

Nonlinear Hamiltonian analysis of the new quadratic torsion theories I. Cases with curvature-free constraints

W.E.V. Barker,[1,2,*] A.N. Lasenby,[1,2,†] M.P. Hobson,[1,‡] and W.J. Handley[1,2,§]

[1] *Astrophysics Group, Cavendish Laboratory, JJ Thomson Avenue, Cambridge CB3 0HE, UK*

[2] *Kavli Institute for Cosmology, Madingley Road, Cambridge CB3 0HA, UK*

When linearised in the absence of matter, 58 cases of the general theory with quadratic curvature and torsion are (i) free from ghosts and tachyons and (ii) power-counting renormalisable. We inspect the nonlinear Hamiltonian structure of the eight cases whose primary constraints do not depend on the curvature tensor. These theories contain a spin-parity 0^- mode: we show that in the nonlinear regime this pseudoscalar is conformally coupled to gravity. We confirm the particle spectrum and classical stability of all these theories in the linear regime. We uncover qualitative dynamical changes in the nonlinear regime, suggesting the activation of ghostly spin-parity sectors and (possible) acausal behaviour.

PACS numbers: 04.50.Kd, 04.60.-m, 04.20.Fy, 98.80.-k

I. INTRODUCTION

For many good reasons, the preferred effective theory of gravity is that of Einstein and Hilbert

$$L_{\mathrm{T}} = -\frac{1}{2}{m_{\mathrm{p}}}^2 R + L_{\mathrm{M}}. \tag{1}$$

This classical theory is powered by the scalar part of the Riemann curvature tensor $R \equiv R^{\mu\nu}{}_{\mu\nu}$, which is the de facto gravitational field strength and contains *second* derivatives of the metric gravitational potential $R \sim \partial^2 g + (\partial g)^2$. Note that the gravitational Lagrangian is related to the total and matter Lagrangia by $L_{\mathrm{G}} \equiv L_{\mathrm{T}} - L_{\mathrm{M}}$, and minimal coupling is assumed in L_{M}.

Two approaches to generalising (1) have proven especially popular

1. The artificially imposed symmetry of the Levi–Civita connection could be relaxed.
2. Higher-order geometric invariants could be added to the Lagrangian.

The first approach leads to a non-vanishing torsion $\mathcal{T}^i_{jk}$, and corresponding non-Riemann curvature $\mathcal{R}^i{}_{jkl}$. The Roman indices refer to a local Lorentz basis, mediated by tetrads (vierbein), or equivalent translational gauge fields $b^i{}_\mu$. The now independent *spin* connection may likewise be cast as a rotational gauge field $A^{ij}{}_\mu$. In terms of these new potentials, the gravitational field strengths $\mathcal{T} \sim \partial b + bA$ and $\mathcal{R} \sim \partial A + A^2$ are closer to the Yang–Mills form familiar from the strong and electroweak sectors of the standard model: they are linear in *first* derivatives and the structure constants of the Poincaré group. By demanding positive parity and freedom from Ostrogradsky ghosts in combination with the second approach, one arrives at the general quadratic $L_{\mathrm{G}} \sim {m_{\mathrm{p}}}^2\mathcal{R} + \mathcal{R}^2 + {m_{\mathrm{p}}}^2\mathcal{T}^2$ theory

$$\begin{aligned} L_{\mathrm{T}} = &-\frac{1}{2}\alpha_0 {m_{\mathrm{p}}}^2\mathcal{R} + {m_{\mathrm{p}}}^2\mathcal{T}_{ijk}\left(\beta_1\mathcal{T}^{ijk} + \beta_2\mathcal{T}^{jik}\right) \\ &+ \beta_3 {m_{\mathrm{p}}}^2\mathcal{T}_i\,\mathcal{T}^i + \alpha_1\mathcal{R}^2 + \mathcal{R}_{ij}\left(\alpha_2\mathcal{R}^{ij} + \alpha_3\mathcal{R}^{ji}\right) \\ &+ \mathcal{R}_{ijkl}\left(\alpha_4\mathcal{R}^{ijkl} + \alpha_5\mathcal{R}^{ikjl} + \alpha_6\mathcal{R}^{klij}\right) + L_{\mathrm{M}}, \end{aligned} \tag{2}$$

where $\mathcal{R}_{ij} \equiv \mathcal{R}^l{}_{ilk}$, $\mathcal{R} \equiv \mathcal{R}^l{}_l$ and $\mathcal{T}_i \equiv \mathcal{T}^l_{il}$. These quadratic terms are added to the scalar curvature invariant for a total of ten dimensionless couplings in the theory. Note that no scalar invariant can be formed from the torsion.

The theory (12) has been deeply studied over four decades. When linearised on a Minkowski background, the theory predicts six massive *roton* (i.e. torsion mode, also known as a *tordion*) modes of spin-parity $J^P = 0^\pm, 1^\pm, 2^\pm$, in addition to the 2^+ mode of the massless graviton. Following early studies by Neville [1, 2], Sezgin and van Nieuwenhuizen found 12 cases of the theory whose propagator poles have positive residues and real masses, i.e. unitary theories [3, 4]. An exhaustive survey by Lin, Hobson, and Lasenby [5, 6] recently found that there are 450 unitary cases in total. Of these, 58 are also power-counting renormalisable (PCR), such that the graviton and roton propagators tend to p^{-4} and p^{-2} respectively in the ultraviolet limit[1].

Unfortunately, the restriction from unitary to PCR-unitary cases of (12) universally switches off the Einstein–Hilbert term

$$\alpha_0 = 0. \tag{3}$$

In the context of (12), this term in isolation constitutes the Einstein–Cartan theory of gravity. Einstein–Cartan theory is dynamically equivalent to GR when

* wb263@cam.ac.uk
† a.n.lasenby@mrao.cam.ac.uk
‡ mph@mrao.cam.ac.uk
§ wh260@cam.ac.uk

[1] With the exception of four cases, some modes violate these rules, but these decouple at high energies without producing divergent loops [6].

the spin tensor of the matter sector vanishes; the linearised theory contains only the 2^+ graviton. Usually, the quadratic $\mathcal{R}^2 + {m_\mathrm{p}}^2\mathcal{T}^2$ terms are viewed as corrections to the scalar ${m_\mathrm{p}}^2\mathcal{R}$, which may be motivated by analogy to Einstein's theory at one loop, $L_\mathrm{G} \sim {m_\mathrm{p}}^2 R + R^2$. A pure $L_\mathrm{G} \sim \mathcal{R}^2 + {m_\mathrm{p}}^2\mathcal{T}^2$ theory is hard to reconcile with this picture, and necessitates a great deal of work since one cannot appeal to the viable Einstein–Cartan limit at low energies. In some sense, a purely quadratic gravity is actually quite natural. While Einstein gravity ammounts to a $\mathbb{R}^{1,3}$ gauge theory of diffeomorphisms, whose gauge potential is $g_{\mu\nu}$, the fields $b^i{}_\mu$ and $A^{ij}{}_\mu$ additionally gauge rotations, and so the whole Poincaré group $\mathbb{R}^{1,3} \rtimes \mathrm{SO}^+(1,3)$. The theory (12) is more properly known as the *quadraic, parity-preserving Poincaré gauge theory* ($\mathrm{PGT}^{\mathrm{q}+}$) of gravity. In this context, a Lagrangian quadratic in Yang–Mills field strengths would make an appealing addition to the standard model – should it prove viable in the nonlinear regime.

The purpose of this series is to test the nonlinear viability of the 58 novel theories by probing their Hamiltonian structure. As a high-spin gauge theory, the $\mathrm{PGT}^{\mathrm{q}+}$ (12) is always singular. The degeneracy of the kinetic Hessian greatly complicates the Lagrangian analysis, incentivising the Hamiltonian approach. By implementing the algorithm of Dirac and Bergmann, we are guaranteed to obtain all propagating degrees of freedom (D.o.F), along with all constraints [7]. In the linearised theory, this is especially easy, and allows us to verify the particle spectra and unitarity of the cases obtained in [5, 6]. In the nonlinear case, the algorithm allows us to flag potentially fatal pathologies which develop under significant departures from Minkowski spacetime – if this spacetime is taken as a vacuum, then the nonlinear regime is equivalent to that of strong fields. In particular, we rely on the simple test of modified gravity set out by Chen, Nester and Yo: *the number and type of constraints should not change in passing from the linear to nonlinear regimes* [8, 9]. The motivation for this criterion is twofold. Generically, a decrease in the number of constraints involves the activation of potentially *ghostly* fields [9]. Moreover, it may be that the nonlinear constraint structure is itself field dependent: this is thought to be associated with the propagation of *acausal* degrees of freedom [8].

In this paper we will test the eight cases in Table II, which are most conducive to the Hamiltonian analysis. Specifically, these are the only cases whose primary constraints are not functions of the curvature. To our knowledge, this practical restriction does no more than to ease the evaluation of commutators. We therefore tentatively view the eight cases to be an *unbiased sample* of the 58 novel theories.

All eight cases fail the prescribed strong-field tests. In some sense, they do so more dramatically than those 'minimal' cases which were previously tested [9], due to the vanishing of mass parameters. Based on these results, we find no evidence that the imposition of the weak-field PCR criteria remedies the chronic ill health of the theory in the strong-field regime. If these findings turn out to be general, it would seem more efficient to perform future surveys in the strong-field regime *from the outset*.

TABLE I. Nonstandard abbreviations.

D.o.F	degrees of freedom
PPM	primary Poisson matrix
P/S/TiC	primary/secondary/tertiary if-constraint
F/SC	first/second-class
iP/S/TF/SC	P/S/TiC which is F/SC on the final shell
sP/SFC	sure primary/secondary constraint, always FC

We are also able to rule the cases out on cosmological grounds, using the scalar-tensor analogue of the background cosmology of the general ten-parameter theory [10]. Two of the eight cases propagate massless modes consistent with long-range gravitational forces, yet their nonlinear cosmological equations are indeterminate. Moreover, we show that these cases are the degenerate limit of an otherwise viable and interesting class of torsion theories in (12), whose background cosmology perfectly replicates that of Einstein's torsion-free gravity (1), conformally coupled to a scalar inflaton ξ

$$L_\mathrm{T} = -\frac{1}{2}{m_\mathrm{p}}^2 R + \frac{1}{12}\xi^2 R + X^{\xi\xi} - \frac{1}{2}{m_\xi}^2\xi^2 + L_\mathrm{M}. \quad (4)$$

Here, the inflaton has kinetic term $X^{\xi\xi} \equiv \frac{1}{2}\nabla_\mu\xi\nabla^\mu\xi$ and mass m_ξ. These results, while fatal to the cases at hand, certainly are not a generic feature of PCR. Indeed, one of the 58 cases has an *excellent* cosmological background [10, 11], though an analysis of its Hamiltonian structure is deferred to the companion paper, since its primary constraints depend on curvature.

Despite our concerns about the strong-field regime, we are able to confirm the weak-field unitarity of all eight cases, and obtain linearised dynamics which are consistent with the particle spectra found in [5, 6].

The remainder of this paper is set out as follows. In Section II we develop the Hamiltonian formulation of the ten-parameter theory (12). In Sections III and IV we apply the Dirac–Bergmann algorithm to each of the linearised cases, and compare with the constraint structure of the nonlinear theories. Conclusions follow in Section VI. Following the conventions of [12] we will use Roman and Greek indices from the middle of the alphabet i, $j\ldots$ μ, $\nu\ldots$ to refer to general Lorentz and coordinate indices running from 0 to three, while a, $b\ldots$ and α, $\beta\ldots$ strictly run from one to three. We use the 'West Coast' signature $(+,-,-,-)$. Our potentially nonstandard acronyms are detailed in Table I.

TABLE II. Select eight of the 58 critical cases of (12), which are unitary and power-counting renormalisable in the absence of matter and when linearised on a Minkowski background. The numbering follows [6], with original numbering of cases first identified in [5] indicated by (∗). The methods of [5, 6] provide partial information about the particle spectrum. Propagators may be of $A^{ij}{}_{\mu}$ (blue), antisymmetric (green) or symmetric (red) parts of $b^{i}{}_{\mu}$. Propagator poles are massless (empty circles) or massive (filled circles). Within each J^P sector, $A^{ij}{}_{\mu}$ and $b^{i}{}_{\mu}$ modes may be coupled (multiple colours) or transmuted by gauge transformations (multiple circles). The final propagating degrees of freedom are shown in the final column, but their field character is indeterminate in the massless case.

#	criticality equalities	unitary inequalities	0^-	0^+	1^-	1^+	2^-	2^+	D.o.F
Case 3	$\hat{\alpha}_1 = \hat{\alpha}_2 = \hat{\alpha}_4 = \hat{\alpha}_6 = \hat{\beta}_2 = \hat{\beta}_1 + 2\hat{\beta}_3 = 0$	$\hat{\alpha}_3 < 0 \wedge \hat{\alpha}_5 < 0 \wedge \hat{\beta}_1 < 0$	●		○ ○		○	○	● ○ ○
Case 17	$\hat{\alpha}_1 = \hat{\alpha}_2 = \hat{\alpha}_3 = \hat{\alpha}_4 = \hat{\alpha}_6 = \hat{\beta}_2 = \hat{\beta}_1 + 2\hat{\beta}_3 = 0$	$\hat{\alpha}_5 < 0$	○		○ ○		○	○	○ ○
Case 20	$\hat{\alpha}_1 = \hat{\alpha}_2 = \hat{\alpha}_4 = \hat{\alpha}_5 = \hat{\alpha}_6 = 0$	$0 < \hat{\beta}_3 \wedge \hat{\alpha}_3 < 0$	●	○	○ ○	○	○	○	
Case 24	$\hat{\alpha}_1 = \hat{\alpha}_2 = \hat{\alpha}_4 = \hat{\alpha}_6 = \hat{\beta}_1 = 0$	$0 < \hat{\beta}_3 \wedge \hat{\alpha}_3 < 0$	●	○	○ ○	○			
Case *5 25	$\hat{\alpha}_1 = \hat{\alpha}_2 = \hat{\alpha}_4 = \hat{\alpha}_5 = \hat{\alpha}_6 = \hat{\beta}_1 = 0$	$0 < \hat{\beta}_3 \wedge \hat{\alpha}_3 < 0$	●	○	○ ○	○			●
Case *6 26	$\hat{\alpha}_1 = \hat{\alpha}_2 = \hat{\alpha}_4 = \hat{\alpha}_5 = \hat{\alpha}_6 = \hat{\beta}_1 = \hat{\beta}_2 = 0$	$0 < \hat{\beta}_3 \wedge \hat{\alpha}_3 < 0$	●			○			
Case 28	$\hat{\alpha}_1 = \hat{\alpha}_2 = \hat{\alpha}_4 = \hat{\alpha}_6 = \hat{\beta}_1 = \hat{\beta}_2 = 0$	$0 < \hat{\beta}_3 \wedge \hat{\alpha}_3 < 0$	●		○	○			
Case 32	$\hat{\alpha}_1 = \hat{\alpha}_2 = \hat{\alpha}_4 = \hat{\alpha}_5 = \hat{\alpha}_6 = \hat{\beta}_2 = 0$	$0 < \hat{\beta}_3 \wedge \hat{\alpha}_3 < 0$	●		○ ○	○	○	○	

II. CONSTRAINED HAMILTONIAN

A. Gauge theory formulation

Recall that in Einstein's theory, the covariant derivative acting on a vector V^μ is simply

$$\nabla_\nu V^\mu \equiv \partial_\nu V^\mu + \Gamma^\mu{}_{\lambda\nu} V^\lambda, \tag{5}$$

where $\Gamma^\mu{}_{\nu\lambda} \equiv \frac{1}{2} g^{\mu\sigma}(\partial_\nu g_{\lambda\sigma} + \partial_\lambda g_{\nu\sigma} - \partial_\sigma g_{\nu\lambda})$ is the Levi–Civita connection with respect to some metric, defined by tangent vectors of the coordinate functions $g_{\mu\nu} \equiv \boldsymbol{e}_\mu \cdot \boldsymbol{e}_\nu$, on a *curved* manifold $\mathcal{M}$. The Riemann curvature tensor is then given by

$$R_{\alpha\beta\mu}{}^{\nu} \equiv 2(\partial_{[\beta}\Gamma^\nu{}_{\alpha]\mu} + \Gamma^\lambda{}_{[\alpha|\mu}\Gamma^\nu{}_{|\beta]\lambda}). \tag{6}$$

This geometric interpretation of gravity is not strictly necessary. In generalising to theories with both curvature and torsion, we adopt the 'particle physics' setup, where the underlying manifold is actually flat Minkowski space $\check{\mathcal{M}}$. The metric for generally curvillinear coordinates is then $\gamma_{\mu\nu} \equiv \boldsymbol{e}_\mu \cdot \boldsymbol{e}_\nu$, and we note that this metric is strictly flat in the sense of (6). Besides the coordinate basis, we define a Lorentz basis whose inner product always returns the Minkowski metric components $\eta_{ij} \equiv \hat{\boldsymbol{e}}_i \cdot \hat{\boldsymbol{e}}_j$. Under this condition, the Lorentz basis is completely free to rotate under the proper, orthochronous Lorentz rotations at each point in $\check{\mathcal{M}}$, and is not presumed to follow from any particular coordinates (i.e. it is non-holonomic). A vector $\mathcal{V}^i$ referred to this basis has a covariant derivative

$$\mathcal{D}_j \mathcal{V}^i \equiv h_j{}^\mu(\partial_\mu \mathcal{V}^i - A_k{}^i{}_\mu \mathcal{V}^k), \tag{7}$$

in which the (inverse) translational gauge field $h_i{}^\mu$ and rotational gauge field $A^{ij}{}_\mu \equiv A^{[ij]}{}_\mu$ are introduced to maintain invariance under general coordinate transformations on $\check{\mathcal{M}}$ (i.e. passively interpreted diffeomorphisms) and rotations of the Lorentz basis. In this way, the Poincaré group is gauged. The inverse translational gauge field satisfies $b^i{}_\mu h_i{}^\nu \equiv \delta^\nu_\mu$ and $b^i{}_\mu h_j{}^\mu \equiv \delta^i_j$. The metric components in the curved space of Einstein's theory (such as the flat cosmological metric we will consider in Section V) can be recovered using $g_{\mu\nu} \equiv \eta_{ij} b^i{}_\mu b^j{}_\nu$ and $g^{\mu\nu} \equiv \eta^{ij} h_i{}^\mu h_j{}^\nu$. The gauge invariant measures on $\mathcal{M}$ and $\check{\mathcal{M}}$ are respectively $\sqrt{-g}$, where $g \equiv \det g_{\mu\nu}$, and $b \equiv h^{-1} \equiv \det b^i{}_\mu$. Two field strength tensors are motivated by commuting the derivative (7)

$$\mathcal{R}^{ij}{}_{kl} \equiv 2h_k{}^\mu h_l{}^\nu \big(\partial_{[\mu} A^{ij}{}_{\nu]} + A^i{}_{o[\mu} A^{oj}{}_{\beta]}\big), \tag{8a}$$

$$\mathcal{T}^i_{kl} \equiv 2h_k{}^\mu h_l{}^\nu \big(\partial_{[\mu} b^i{}_{\nu]} + A^i{}_{o[\mu} b^o{}_{\nu]}\big). \tag{8b}$$

These are the Riemann–Cartan curvature and torsion tensors, but are not understood to imbue $\check{\mathcal{M}}$ with any geometry. While the flat space picture is consistent with our earlier treatments on this topic [10, 11, 13], we concede that it is far more usual to treat torsion as a geometric quality, complimentary to curvature. In the geometric picture, the Riemann spacetime $\mathcal{M}$ is generalised to a Riemann–Cartan spacetime, rather than specialised to a Minkowski spacetime. The interpretations are dynamically indistinguishable, and translations between the two are provided in [12, 13].

B. Primary constraints and $3+1$

In order to transition to the Hamiltonian picture [7, 12, 14], we first define the *canonical* momenta from the Lagrangian

$$\pi_i{}^\mu \equiv \frac{\partial b L_{\mathrm{T}}}{\partial \partial_0 b^i{}_\mu}, \quad \pi_{ij}{}^\mu \equiv \frac{\partial b L_{\mathrm{T}}}{\partial \partial_0 A^{ij}{}_\mu}. \tag{9}$$

Since the field strengths (8a) and (8b) from which (2) is constructed make no reference to the velocities of $b^k{}_0$ and $A^{ij}{}_0$, the definitions (9) incur 10 primary constraints

$$\varphi_k{}^0 \equiv \pi_k{}^0 \approx 0, \quad \varphi_{ij}{}^0 \equiv \pi_{ij}{}^0 \approx 0, \tag{10}$$

so that the conjugate fields $b^k{}_0$ and $A^{ij}{}_0$ are non-physical. Notice that the *weak* equality is denoted by ($\approx$). The constraints (10) are a consequence of Poincaré gauge symmetry; their presence is independent of the couplings, and they are first-class (FC). We refer to them as 'sure' primary first class (sPFC) constraints. In order to systematically isolate the 'sure' non-physical fields, we introduce the 3+1 (ADM) splitting of the spacetime, in which a spacelike foliation is characterised by timelike unit normal n_k. Any vector which refers to the local Lorentz basis may be split into perpendicular and parallel components $\mathcal{V}^i = \mathcal{V}^\perp n^i + \mathcal{V}^{\bar{i}}$ with respect to this normal: parallel indices are always denoted with an overbar. In what follows, it is very useful to note the identities $b^{\bar{k}}{}_\alpha h_{\bar{l}}{}^\alpha = \delta^{\bar{k}}_{\bar{l}}$ and $b^{\bar{k}}{}_\alpha h_{\bar{k}}{}^\beta = \delta^\beta_\alpha$. The *lapse* function and *shift* vector are defined with reference to the non-physical part of the translational gauge field using this normal

$$N \equiv n_k b^k{}_0, \quad N^\alpha \equiv h_{\bar{k}}{}^\alpha b^k{}_0. \tag{11}$$

The remaining momenta can be expressed in the 'parallel' form $\hat{\pi}_i{}^{\bar{k}} \equiv \pi_i{}^\alpha b^k{}_\alpha$ and $\hat{\pi}_{ij}{}^{\bar{k}} \equiv \pi_{ij}{}^\alpha b^k{}_\alpha$. In order to reveal the Hamiltonian structure of the theory, the Lagrangian in (2) is best written in the irreducible form

$$\begin{aligned} L_{\mathrm{T}} = -\frac{1}{2}\hat{\alpha}_0 m_{\mathrm{p}}{}^2 \mathcal{R} + \sum_{I=1}^{6} \hat{\alpha}_I \mathcal{R}^{ij}{}_{kl}\, {}^I\mathcal{P}_{ij}{}^{kl}{}_{nm}{}^{pq} \mathcal{R}^{nm}{}_{pq} \\ + m_{\mathrm{p}}{}^2 \sum_{I=1}^{3} \hat{\beta}_I \mathcal{T}^i_{jk}\, {}^I\mathcal{P}_i{}^{jk}{}_l{}^{nm} \mathcal{T}^l_{nm} + L_{\mathrm{M}}. \end{aligned} \tag{12}$$

where the nine operators ${}^I\mathcal{P}^{\cdots}_{\cdots}$ project out all the irreducible representations of $\mathrm{SO}(1,3)$, in no particaular order, from the field strengths. Our original naïve couplings in (2) are expressible in terms of their superior irreducible counterparts according to $\alpha_0 \equiv \hat{\alpha}_0$ and

$$\begin{aligned} \alpha_1 &\equiv \hat{\alpha}_4 + \frac{1}{2}\hat{\alpha}_5 + \hat{\alpha}_6, \quad \alpha_2 \equiv \hat{\alpha}_4 - \hat{\alpha}_6, \\ \alpha_3 &\equiv \hat{\alpha}_4 - \hat{\alpha}_5 + \hat{\alpha}_6, \\ \alpha_4 &\equiv \frac{1}{2}\hat{\alpha}_2 + \frac{1}{2}\hat{\alpha}_3 + \hat{\alpha}_4 + \frac{1}{2}\hat{\alpha}_5 + \hat{\alpha}_6, \\ \alpha_5 &\equiv \frac{1}{2}\hat{\alpha}_2 - \frac{1}{2}\hat{\alpha}_3 + \hat{\alpha}_4 - \hat{\alpha}_6, \\ \alpha_6 &\equiv 6\hat{\alpha}_1 + \frac{3}{2}\hat{\alpha}_2 + \frac{3}{2}\hat{\alpha}_3 + \hat{\alpha}_4 + \frac{1}{2}\hat{\alpha}_5 + \hat{\alpha}_6, \\ \beta_1 &\equiv \hat{\beta}_1 + \frac{1}{2}\hat{\beta}_2, \quad \beta_2 \equiv \hat{\beta}_1 + \frac{1}{2}\hat{\beta}_2 + \frac{3}{2}\hat{\beta}_3, \\ \beta_3 &\equiv \hat{\beta}_1 - \hat{\beta}_2. \end{aligned} \tag{13}$$

Within the field strengths, the $3+1$ splitting is used again to separate out the 'sure' non-physical fields and non-canonical velocities

$$\mathcal{T}^i_{kl} = \mathcal{T}^i_{\bar{k}\bar{l}} + 2n_{[k}\mathcal{T}^i_{\perp\bar{l}]}\,, \tag{14a}$$

$$\mathcal{R}^{ij}{}_{kl} = \mathcal{R}^{ij}{}_{\bar{k}\bar{l}} + 2n_{[k}\mathcal{R}^{ij}{}_{\perp\bar{l}]}, \tag{14b}$$

where such variables are confined to the second term in each case. We are concerned with theories of the quadratic $L_{\mathrm{G}} \sim \mathcal{R}^2 + m_{\mathrm{p}}{}^2\mathcal{T}^2$ form (i.e. $\hat{\alpha}_0 = 0$), under the source-free condition $L_{\mathrm{M}} = 0$. Substituting (12) into (9) and using Eqs. (14a) and (14b), we find that the parallel momenta can be neatly expressed as functions of the field strengths

$$\frac{\hat{\pi}_i{}^{\bar{k}}}{J} = \frac{\partial L_{\mathrm{T}}}{\partial \mathcal{T}^i_{\perp\bar{k}}} = 4m_{\mathrm{p}}{}^2 \sum_{I=1}^{3} \hat{\beta}_I\, {}^I\mathcal{P}_i{}^{\perp\bar{k}}{}_n{}^{mo} \mathcal{T}^n{}_{mo}\,, \tag{15a}$$

$$\frac{\hat{\pi}_{ij}{}^{\bar{k}}}{J} = \frac{\partial L_{\mathrm{T}}}{\partial \mathcal{R}^{ij}{}_{\perp\bar{k}}} = 8\sum_{I=1}^{6} \hat{\alpha}_I\, {}^I\mathcal{P}_{ij}{}^{\perp\bar{k}}{}_{mn}{}^{pq} \mathcal{R}^{mn}{}_{pq}, \tag{15b}$$

where the measure $J = b/N$ on the foliation is strictly physical, making no reference to $b^k{}_0$.

Writing the parallel momenta in this form facilitates the identification of further primary constraints. Beginning with (15a), we find that the 12 translational parallel momenta decompose into four irreducible representations of O(3). Using the spin-parity notation of [9, 15] we write these as

$$\hat{\pi}_{k\bar{l}} = \hat{\pi}_{\bar{k}\bar{l}} + n_k \hat{\pi}_{\perp\bar{l}}, \tag{16a}$$

$$\hat{\pi}_{\bar{k}\bar{l}} = \frac{1}{3}\eta_{\bar{k}\bar{l}}\hat{\pi} + \hat{\hat{\pi}}_{\bar{k}\bar{l}} + \tilde{\hat{\pi}}_{\bar{k}\bar{l}}, \tag{16b}$$

and note that they are the 0^+, 1^+, 1^- and 2^- sectors. Applying this decomposition to (15a) as a whole, we obtain four functions which, with the aid of (14a), are simultaneously defined both in terms of canonical and non-canonical variables

$$\varphi \equiv J^{-1}\hat{\pi} = \hat{\beta}_2 m_{\mathrm{p}}{}^2 \eta^{\bar{k}\bar{l}} \mathcal{T}_{\bar{k}\bar{l}\perp}\,, \tag{17a}$$

$$\begin{aligned} \hat{\varphi}_{\bar{k}\bar{l}} &\equiv J^{-1}\hat{\hat{\pi}}_{\bar{k}\bar{l}} - \frac{4}{3}(\hat{\beta}_1 - \hat{\beta}_3) m_{\mathrm{p}}{}^2 \mathcal{T}_{\perp\bar{k}\bar{l}} \\ &= \frac{1}{3}(\hat{\beta}_1 + 2\hat{\beta}_3) m_{\mathrm{p}}{}^2 \mathcal{T}_{[\bar{k}\bar{l}]\perp}\,, \end{aligned} \tag{17b}$$

$$\begin{aligned} \varphi_{\perp\bar{k}} &\equiv J^{-1}\hat{\pi}_{\perp\bar{k}} - \frac{4}{3}(\hat{\beta}_1 - \hat{\beta}_2) m_{\mathrm{p}}{}^2 \vec{\mathcal{T}}_{\bar{k}} \\ &= \frac{1}{3}(2\hat{\beta}_1 + \hat{\beta}_2) m_{\mathrm{p}}{}^2 \mathcal{T}_{\perp\bar{k}\perp}\,, \end{aligned} \tag{17c}$$

$$\tilde{\varphi}_{\bar{k}\bar{l}} \equiv J^{-1}\tilde{\hat{\pi}}_{\bar{k}\bar{l}} = \hat{\beta}_1 m_{\mathrm{p}}{}^2 \mathcal{T}_{\langle\bar{k}\bar{l}\rangle\perp}\,, \tag{17d}$$

where the vector and traceless, symmetric torsion are

$$\vec{\mathcal{T}}_{\bar{k}} \equiv \mathcal{T}^{\bar{i}}_{\bar{k}\bar{i}}\,, \quad \mathcal{T}_{\langle\bar{k}\bar{l}\rangle\perp} \equiv \mathcal{T}_{(\bar{k}\bar{l})\perp} - \tfrac{1}{3}\eta_{\bar{k}\bar{l}}\eta^{\bar{i}\bar{j}}\mathcal{T}_{\bar{i}\bar{j}\perp}\,. \tag{18}$$

In each case, *if* the combination of coupling constants appearing in the non-canonical RHS definition vanishes, the canonically defined function on the LHS becomes a

primary 'if' constraint (PiC). An analogous construction is available for (15b), with the 18 remaining momenta decomposing as follows

$$\hat{\pi}_{kl\overline{o}} = \hat{\pi}_{\overline{klo}} + 2n_{[k}\hat{\pi}_{\perp\overline{l}]\overline{o}}, \tag{19a}$$

$$\hat{\pi}_{\perp\overline{kl}} = \frac{1}{3}\eta_{\overline{kl}}\hat{\pi}_{\perp} + \overset{\wedge}{\hat{\pi}}_{\perp\overline{kl}} + \overset{\sim}{\hat{\pi}}_{\perp\overline{kl}}, \tag{19b}$$

$$\hat{\pi}_{\overline{klo}} = \frac{1}{6}\epsilon_{\overline{klo}\perp}{}^{\mathrm{P}}\hat{\pi} + \overset{\rightharpoonup}{\hat{\pi}}_{[\overline{k}}\eta_{\overline{l}]\overline{o}} + \frac{4}{3}{}^{\mathrm{T}}\hat{\pi}_{\overline{klo}}. \tag{19c}$$

These are the 0^+, 1^+ and 2^+, and then 0^-, 1^- and 2^- representations. The six PiC functions from (15b) are then

$$\begin{aligned}\varphi_{\perp} &\equiv J^{-1}\hat{\pi}_{\perp} + 2\left(\hat{\alpha}_4 - \hat{\alpha}_6\right)\underline{\mathcal{R}} \\ &= \left(\hat{\alpha}_4 + \hat{\alpha}_6\right)\mathcal{R}_{\perp\perp},\end{aligned} \tag{20a}$$

$$\begin{aligned}{}^{\mathrm{P}}\varphi &\equiv J^{-1}{}^{\mathrm{P}}\hat{\pi} + 4\left(\hat{\alpha}_2 - \hat{\alpha}_3\right){}^{\mathrm{P}}\mathcal{R}_{\perp\circ} \\ &= \left(\hat{\alpha}_2 + \hat{\alpha}_3\right){}^{\mathrm{P}}\mathcal{R}_{\circ\perp},\end{aligned} \tag{20b}$$

$$\begin{aligned}\overset{\wedge}{\varphi}_{\perp\overline{kl}} &\equiv J^{-1}\overset{\wedge}{\hat{\pi}}_{\perp\overline{kl}} + 4\left(\hat{\alpha}_2 - \hat{\alpha}_5\right)\underline{\mathcal{R}}_{[\overline{kl}]} \\ &= \left(\hat{\alpha}_2 + \hat{\alpha}_5\right)\mathcal{R}_{\perp[\overline{kl}]\perp},\end{aligned} \tag{20c}$$

$$\begin{aligned}\overset{\rightharpoonup}{\varphi}_{\overline{k}} &\equiv J^{-1}\overset{\rightharpoonup}{\hat{\pi}}_{\overline{k}} + 4\left(\hat{\alpha}_4 - \hat{\alpha}_5\right)\mathcal{R}_{\perp\overline{k}} \\ &= \left(\hat{\alpha}_4 + \hat{\alpha}_5\right)\mathcal{R}_{\overline{k}\perp},\end{aligned} \tag{20d}$$

$$\begin{aligned}\overset{\sim}{\varphi}_{\perp\overline{kl}} &\equiv J^{-1}\overset{\sim}{\hat{\pi}}_{\perp\overline{kl}} + 4\left(\hat{\alpha}_1 - \hat{\alpha}_4\right)\underline{\mathcal{R}}_{\langle\overline{kl}\rangle} \\ &= \left(\hat{\alpha}_1 + \hat{\alpha}_4\right)\mathcal{R}_{\perp\langle\overline{kl}\rangle\perp},\end{aligned} \tag{20e}$$

$$\begin{aligned}{}^{\mathrm{T}}\varphi_{\overline{klo}} &\equiv J^{-1}{}^{\mathrm{T}}\hat{\pi}_{\overline{klo}} - 4\left(\hat{\alpha}_1 - \hat{\alpha}_2\right){}^{\mathrm{T}}\mathcal{R}_{\perp\overline{klo}} \\ &= \left(\hat{\alpha}_1 + \hat{\alpha}_2\right){}^{\mathrm{T}}\mathcal{R}_{\overline{klo}\perp},\end{aligned} \tag{20f}$$

where we make further notational definitions

$$\begin{gathered}{}^{\mathrm{P}}\mathcal{R}_{\perp\circ} \equiv \epsilon^{\overline{ijk}\perp}\mathcal{R}_{\overline{ijk}\perp}, \quad {}^{\mathrm{P}}\mathcal{R}_{\circ\perp} \equiv \epsilon^{\overline{ijk}\perp}\mathcal{R}_{\perp\overline{ijk}}, \\ \underline{\mathcal{R}}_{\overline{kl}} \equiv \mathcal{R}^{\overline{i}}{}_{\overline{kli}}, \quad \underline{\mathcal{R}} \equiv \underline{\mathcal{R}}^{\overline{i}}{}_{\overline{i}}.\end{gathered} \tag{21}$$

By this analysis, the possible occurance of primary constraints is systematically exhausted.

C. Secondary constraints and the Hamiltonian

In order to be consistent, a primary constraint should not have any velocity within the final mass shell, so its commutator with the *total* Hamiltonian should weakly vanish

$$\dot{\varphi}(x_1) \equiv \int \mathrm{d}^3x_2\Big\{\varphi(x_1), \mathcal{H}_{\mathrm{T}}(x_2)\Big\} \approx 0. \tag{22}$$

The the total Hamiltonian is related to the *canonical* Hamiltonian, the Legendre-transformed Lagrangian, by the constraints and their multiplier fields

$$\mathcal{H}_{\mathrm{T}} \equiv \mathcal{H}_{\mathrm{C}} + u^k{}_0\varphi_k{}^0 + \frac{1}{2}u^{ij}{}_0\varphi_{ij}{}^0 + (u\cdot\varphi), \tag{23}$$

where the last term schematically represents any PiCs which may arise. The canonical Hamiltonian may generally be collected into the insightful *Dirac* form

$$\mathcal{H}_{\mathrm{C}} \equiv N\mathcal{H}_{\perp} + N^{\alpha}\mathcal{H}_{\alpha} - \frac{1}{2}A^{ij}{}_0\mathcal{H}_{ij} + \partial_{\alpha}\mathscr{D}^{\alpha}, \tag{24}$$

i.e. as a linear function of the non-physical fields up to a surface term. The remaining functions which appear in (24) are defined as follows

$$\mathcal{H}_{\perp} \equiv \hat{\pi}_i{}^{\overline{k}}\mathcal{T}^i{}_{\perp\overline{k}} + \frac{1}{2}\hat{\pi}_{ij}{}^{\overline{k}}\mathcal{R}^{ij}{}_{\perp\overline{k}} - JL - n^kD_{\alpha}\pi_k{}^{\alpha}, \tag{25a}$$

$$\mathcal{H}_{\alpha} \equiv \pi_i{}^{\beta}T^i{}_{\alpha\beta} + \frac{1}{2}\pi_{ij}{}^{\beta}R^{ij}{}_{\alpha\beta} - b^k{}_{\alpha}D_{\beta}\pi_k{}^{\beta}, \tag{25b}$$

$$\mathcal{H}_{ij} \equiv 2\pi_{[i}{}^{\alpha}b_{j]\alpha} - D_{\alpha}\pi_{ij}{}^{\alpha}, \tag{25c}$$

$$\mathscr{D}^{\alpha} \equiv b^i{}_0\pi_i{}^{\alpha} + \frac{1}{2}A^{ij}{}_0\pi_{ij}{}^{\alpha}. \tag{25d}$$

It is clear from (22) and the Dirac form (24) that the consistency of (10) invokes 10 'sure' *secondary* first class (sSFC) constriants

$$\mathcal{H}_{\perp} \approx 0, \quad \mathcal{H}_{\alpha} \approx 0, \quad \mathcal{H}_{ij} \approx 0. \tag{26}$$

It is important to note that while the sSFCs in (26) are always enforced, it does not always follow that 2×10 D.o.F are removed from the theory, as is the case with the sPFCs in (10). The functions involved are quite complicated, and may degenerately express a reduced number of FCs, or FCs which only appear at deeper levels in the consistency chain. Indeed, while this is very rare in the literature, we will find that it occurs for all the novel theories, as a consequence of vanishing mass parameters.

The linear and rotational super-momenta $\mathcal{H}_{\alpha}$ and $\mathcal{H}_{ij}$ are kinematic generators which do not impinge on the dynamics. Thus, in the evaluation of (22), it is sufficient to work purely with the super-Hamiltonian $\mathcal{H}_{\perp}$, which is readily expanded out using Eqs. (12), (14a) and (14b) to give

$$
\begin{aligned}
\mathcal{H}_\perp = & \sum_{I=1}^{3} {m_\mathrm{p}}^2 J\hat{\beta}_I \Big[4\mathcal{T}^i_{\perp\overline{k}}\, {}^I\mathcal{P}_i{}^{\perp\overline{k}}{}_j{}^{\perp\overline{l}}\mathcal{T}^j_{\perp\overline{l}} - \mathcal{T}^i_{\overline{mk}}\, {}^I\mathcal{P}_i{}^{\overline{mk}}{}_j{}^{\overline{nl}}\mathcal{T}^j_{\overline{nl}}\Big] + \sum_{I=1}^{6} J\hat{\alpha}_I \Big[4\mathcal{R}^{ip}{}_{\perp\overline{k}}\, {}^I\mathcal{P}_{ip}{}^{\perp\overline{k}}{}_{jq}{}^{\perp\overline{l}}\mathcal{R}^{jq}{}_{\perp\overline{l}} \\
& - \mathcal{R}^{ip}{}_{\overline{mk}}\, {}^I\mathcal{P}_{ip}{}^{\overline{mk}}{}_{jq}{}^{\overline{nl}}\mathcal{R}^{jq}{}_{\overline{nl}}\Big] - n^k D_\alpha \pi_k{}^\alpha \\
= & \frac{J}{16}\Bigg[\frac{2\varphi^2}{3\hat{\beta}_2 {m_\mathrm{p}}^2} + \frac{6\hat{\varphi}_{\overline{kl}}\hat{\varphi}^{\overline{kl}}}{(\hat{\beta}_1+2\hat{\beta}_3){m_\mathrm{p}}^2} + \frac{6\varphi_{\perp\overline{k}}\varphi^{\perp\overline{k}}}{(2\hat{\beta}_1+\hat{\beta}_2){m_\mathrm{p}}^2} + \frac{2\widetilde{\varphi}_{\overline{kl}}\widetilde{\varphi}^{\overline{kl}}}{\hat{\beta}_1 {m_\mathrm{p}}^2} + \frac{2{\varphi_\perp}^2}{3(\hat{\alpha}_4+\hat{\alpha}_6)} + \frac{{}^\mathrm{P}\varphi^2}{6(\hat{\alpha}_2+\hat{\alpha}_3)} + \frac{2\hat{\varphi}_{\perp\overline{kl}}\hat{\varphi}^{\perp\overline{kl}}}{\hat{\alpha}_2+\hat{\alpha}_5} \\
& + \frac{\vec{\varphi}_{\overline{k}}\vec{\varphi}^{\overline{k}}}{\hat{\alpha}_4+\hat{\alpha}_5} + \frac{2\widetilde{\varphi}_{\perp\overline{kl}}\widetilde{\varphi}^{\perp\overline{kl}}}{\hat{\alpha}_1+\hat{\alpha}_4} + \frac{16\,{}^\mathrm{T}\varphi_{\overline{klo}}\,{}^\mathrm{T}\varphi^{\overline{klo}}}{9(\hat{\alpha}_1+\hat{\alpha}_2)}\Bigg] + J\Bigg[\frac{1}{3}(2\hat{\beta}_1+\hat{\beta}_3){m_\mathrm{p}}^2\mathcal{T}_{\perp\overline{kl}}\,\mathcal{T}^{\perp\overline{kl}} + \frac{1}{3}(\hat{\beta}_1+2\hat{\beta}_2){m_\mathrm{p}}^2\vec{\mathcal{T}}_{\overline{k}}\vec{\mathcal{T}}^{\overline{k}} \\
& - \frac{1}{6}\hat{\beta}_3 {m_\mathrm{p}}^2\,{}^\mathrm{P}\mathcal{T}^2 + \frac{16}{9}\hat{\beta}_1 {m_\mathrm{p}}^2\,{}^\mathrm{T}\mathcal{T}_{\overline{klo}}\,{}^\mathrm{T}\mathcal{T}^{\overline{klo}} + \frac{1}{6}(\hat{\alpha}_4+\hat{\alpha}_6)\underline{\mathcal{R}}^2 - \frac{1}{6}(\hat{\alpha}_2+\hat{\alpha}_3)\,{}^\mathrm{P}\mathcal{R}_{\perp\mathrm{o}}{}^2 + 2(\hat{\alpha}_2+\hat{\alpha}_5)\underline{\mathcal{R}}_{[\overline{kl}]}\underline{\mathcal{R}}^{[\overline{kl}]} \\
& + (\hat{\alpha}_4+\hat{\alpha}_5)\mathcal{R}_{\perp\overline{k}}\mathcal{R}^{\perp\overline{k}} + 2(\hat{\alpha}_1+\hat{\alpha}_4)\underline{\mathcal{R}}_{\langle\overline{kl}\rangle}\underline{\mathcal{R}}^{\langle\overline{kl}\rangle} + \frac{16}{9}(\hat{\alpha}_1+\hat{\alpha}_2)\,{}^\mathrm{T}\mathcal{R}_{\perp\overline{klo}}\,{}^\mathrm{T}\mathcal{R}^{\perp\overline{klo}}\Bigg] - n^k D_\alpha \pi_k{}^\alpha .
\end{aligned}
\tag{27}
$$

To arrive at the second equality in (27), the non-canonical 'perpendicaular' field strengths appearing in the first equality are canonicalised at length by the dual PiC definitions in Eqs. (17a) to (17d) and Eqs. (20a) to (20f), resulting in terms quadratic in the PiC functions, and in the canonical 'parallel' field strengths. The resulting expression is quite lengthy, but can be simplified for any given theory by safely elimiating those PiC functions which become constraints. The signs of the remaining quadratic PiC terms are then instrumental in the identification of unconstrained ghosts, since the PiC functions are schematically of the form

$$
\varphi \sim \pi + \mathcal{R}, \quad \varphi \sim \pi + \mathcal{T}. \tag{28}
$$

The consistency of the PiCs is less straightforward. Generally, the PiCs may be FC or SC within their own mass shell. In the case that a PiC is FC, (22) provides a secondary if-constraint (SiC). Possibly, the PiC and SiC do not commute; in that case both become SC within the new mass shell and the consistency of the SiC allows a multiplier to be determined

$$
\begin{aligned}
\dot{\chi}(x_1) \equiv \int \mathrm{d}^3x_2 \Big(& N\Big\{\chi(x_1), \mathcal{H}_\perp(x_2)\Big\} \\
& + u\cdot\Big\{\chi(x_1), \varphi(x_2)\Big\}\Big) \approx 0.
\end{aligned}
\tag{29}
$$

Otherwise, a tertiary if-constraint (TiC) may be found, and the process continues until the contraint chain from the PiC is absorbed by another chain, or by the sSFCs. In the case that a PiC is already SC within the PiC mass shell, its chain terminates immediately and two multipliers are determined. We note that occasionally, a constraint may be encountered at some deep level which retroactively terminates the chain at a shallower point. Only once the algoritm has terminated is it safe to categorise the if-constraints as FC or SC.

In the linearised theory, the analysis is greatly simplified by an understanding of the mass spectrum. Only the $\mathcal{O}(1)$ parts of the PiC commutators contribute to the evaluation of the multipliers. Such commutators are possible only between pairs of PiCs which belong to the same O(3) irrep, and which are known as *conjugate pairs*. Conjugate PiCs will fail to commute in the linear theory only when their common mass parameter is non-vanishing. In this case, if only one PiC in a pair is present, it will still fail to commute with the SiC that maintains its consistency. Particlularly, the rotational ${}^\mathrm{P}\varphi$ and ${}^\mathrm{T}\varphi_{\overline{klo}}$ have no O(3) counterparts in the translational sector, and are conjugate with their secondaries ${}^\mathrm{P}\chi$ and ${}^\mathrm{T}\chi_{\overline{klo}}$ a priori. In the case of *vanishing* mass parameters, the PiCs are FC, and a new gauge symmetry is invoked. The PiCs belonging to various O(3) irreps, along with their kinetic parameters and linearised mass parameters are listed in Table III.

TABLE III. Spin-parity sectors and associated PiCs, along with their kinetic and mass parameters. For completeness, we include the ${m_\mathrm{p}}^2\mathcal{R}$ term, mediated by $\hat{\alpha}_0$.

J^P	PiC	Kinetic parameter	Mass parameter
0^+	φ	$\hat{\beta}_2$	$\hat{\alpha}_0(2\hat{\alpha}_0+\hat{\beta}_2)$
	$\varphi_\perp$	$\hat{\alpha}_4+\hat{\alpha}_6$	
0^-	${}^\mathrm{P}\varphi$	$\hat{\alpha}_2+\hat{\alpha}_3$	$\hat{\alpha}_0+2\hat{\beta}_3$
1^+	$\hat{\varphi}_{\overline{kl}}$	$\hat{\beta}_1+2\hat{\beta}_3$	$(\hat{\alpha}_0+2\hat{\beta}_3)(\hat{\alpha}_0-\hat{\beta}_1)$
	$\hat{\varphi}_{\perp\overline{kl}}$	$\hat{\alpha}_2+\hat{\alpha}_5$	
1^-	$\varphi_{\perp\overline{k}}$	$2\hat{\beta}_1+\hat{\beta}_2$	$(2\hat{\alpha}_0+\hat{\beta}_2)(\hat{\alpha}_0-\hat{\beta}_1)$
	$\vec{\varphi}_{\overline{k}}$	$\hat{\alpha}_4+\hat{\alpha}_5$	
2^+	$\widetilde{\varphi}_{\overline{kl}}$	$\hat{\beta}_1$	$\hat{\alpha}_0(\hat{\alpha}_0-\hat{\beta}_1)$
	$\widetilde{\varphi}_{\perp\overline{kl}}$	$\hat{\alpha}_1+\hat{\alpha}_4$	
2^-	${}^\mathrm{T}\varphi_{\overline{klo}}$	$\hat{\alpha}_1+\hat{\alpha}_2$	$\hat{\alpha}_0-\hat{\beta}_1$

The evaluation of Poisson brackets is made tedious by the dependence of various quantities on the translational gauge field, as illustrated by the following useful identi-

ties

$$\frac{\partial n_l}{\partial b^k{}_\mu} = -n_k h_{\bar{l}}{}^\mu, \quad \frac{\partial h_l{}^\nu}{\partial b^k{}_\mu} = -h_k{}^\nu h_l{}^\mu,$$
$$\frac{\partial b}{\partial b^k{}_\nu} = b h_k{}^\nu, \quad \frac{\partial J}{\partial b^k{}_\nu} = J h_{\bar{k}}{}^\nu, \quad \frac{\partial N}{\partial b^k{}_\nu} = N n_k h_\perp{}^\nu. \tag{30}$$

As a crude measure to simplify the calculations, we artificially restrict our analysis to theories whose PiCs among Eqs. (20a) to (20f) do not depend on $\mathcal{R}^{ij}{}_{\overline{kl}}$. It must be emphasised that this does not (to our knowledge) translate into any useful restriction on the physics. Of the 58 novel theories in [5, 6], eight satisfy our criterion: Case 3, Case 17, Case 20, Case 24, Case ^{*5}25, Case ^{*6}26, Case 28 and Case 32. We detail in Table II our prior understanding of these theories, as encoded by the saturated graviton and roton propagators, linearised on Minkowski spacetime in the absence of matter. Case 3 and Case 17 are particularly interesting as candidate theories of gravity, as they contain two massless D.o.F with power in the 2^+ part of the propagator – we will return to this point in Section IV.

From our discussion in Section II C, we see that the constraint structure of the theory depends partially on the commutators between the PiCs, which form the *primary Poisson matrix* (PPM). In Sections III and IV we will use the structure of the nonlinear PPM as a proxy for the health of each theory.

III. MASSIVE-ONLY THEORIES

A. Case ^{*6}26

Conveniently, the PiCs of the massive theories do not depend on $\mathcal{T}^i{}_{\overline{jk}}$, as well as $\mathcal{R}^{ij}{}_{\overline{kl}}$, so we will have schematically $\varphi \sim \pi$ for both translational and rotational sectors. By substituting the definition of Case ^{*6}26 from Table II into (27) and (25a), the total Hamiltonian is seen to take the form

$$\mathcal{H}_{\mathrm{T}} = \frac{b}{96}\left(\frac{18\hat{\varphi}_{\overline{kl}}\hat{\varphi}^{\overline{kl}}}{\hat{\beta}_3} - \frac{{}^{\mathrm{P}}\varphi^2}{\hat{\alpha}_3}\right) + \ldots, \tag{31}$$

where we include only the part quadratic in the momenta. The remaining eight PiC functions that do not appear in (31) are primarily constrained, and give rise to the following nonvanishing commutators within the PiC shell

$$\left\{\varphi_{\perp\bar{i}}, \varphi_{\perp\bar{l}}\right\} \approx \frac{2}{J^2}\hat{\hat{\pi}}_{\bar{i}\bar{l}}\delta^3, \tag{32a}$$
$$\left\{\varphi_{\perp\bar{i}}, \hat{\hat{\varphi}}_{\perp\overline{lm}}\right\} \approx -\frac{1}{6J^2}\epsilon_{\bar{i}\overline{lm}\perp}{}^{\mathrm{P}}\hat{\pi}\,\delta^3, \tag{32b}$$
$$\left\{\tilde{\varphi}_{\overline{ij}}, \tilde{\varphi}_{\overline{lm}}\right\} \approx \frac{1}{J^2}\left[\eta_{\bar{i}(\bar{l}|}\hat{\hat{\pi}}_{\bar{j}|\overline{m})} + \eta_{\bar{j}(\bar{l}|}\hat{\hat{\pi}}_{\bar{i}|\overline{m})}\right]\delta^3, \tag{32c}$$
$$\left\{\tilde{\varphi}_{\overline{ij}}, {}^{\mathrm{T}}\varphi_{\overline{lmn}}\right\} \approx \frac{1}{24J^2}\Big[2\eta_{(\bar{i}|\overline{n}}\epsilon_{|\bar{j})\overline{lm}\perp} - \eta_{(\bar{i}|\bar{l}}\epsilon_{|\bar{j})\overline{mn}\perp} + \eta_{(\bar{i}|\overline{m}}\epsilon_{|\bar{j})\overline{ln}\perp}\Big]{}^{\mathrm{P}}\hat{\pi}\,\delta^3, \tag{32d}$$

where δ^3 represents the equal-time Dirac function. The nonlinear PPM of Case ^{*6}26 is then written:

$$\begin{array}{c|cccccccc|c}
 & \varphi & \varphi_{\perp\bar{k}} & \tilde{\varphi}_{\overline{kl}} & \varphi_\perp & \hat{\hat{\varphi}}_{\perp\overline{kl}} & \vec{\varphi}_{\bar{k}} & \tilde{\varphi}_{\perp\overline{kl}} & {}^{\mathrm{T}}\varphi_{\overline{klo}} & \\
\hline
\varphi & \cdot & \cdot & \cdot & \boxed{\cdot} & \cdot & \cdot & \cdot & \cdot & 1 \\
\varphi_{\perp\bar{k}} & \cdot & \hat{\pi} & \cdot & \cdot & \hat{\pi}! & \boxed{\cdot} & \cdot & \cdot & 3 \\
\tilde{\varphi}_{\overline{kl}} & \cdot & \cdot & \hat{\pi} & \cdot & \cdot & \cdot & \boxed{\cdot} & \hat{\pi}! & 5 \\
\varphi_\perp & \boxed{\cdot} & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & 1 \\
\hat{\hat{\varphi}}_{\perp\overline{kl}} & \cdot & \hat{\pi}! & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & 3 \\
\vec{\varphi}_{\bar{k}} & \cdot & \boxed{\cdot} & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & 3 \\
\tilde{\varphi}_{\perp\overline{kl}} & \cdot & \cdot & \boxed{\cdot} & \cdot & \cdot & \cdot & \cdot & \cdot & 5 \\
{}^{\mathrm{T}}\varphi_{\overline{klo}} & \cdot & \cdot & \hat{\pi}! & \cdot & \cdot & \cdot & \cdot & \cdot & 5 \\
\hline
 & 1 & 3 & 5 & 1 & 3 & 3 & 5 & 5 &
\end{array} \tag{33}$$

The elements of the matrix schematically represent the nonlinear Poisson brackets between the PiCs. The PiCs are labelled, along with their multiplicities, at the edges of the PPM. They are arranged so as to divide the matrix into translational and rotational blocks, separated by (+). All brackets are restricted to the PiC shell. Commuting PiCs are dentoted by $(\cdot)$. Non-commuting PiCs denoted as $(\hat{\pi})$ are strictly linear combinations of $\hat{\pi}_{ij}{}^{\bar{k}}$ and $\hat{\pi}_i{}^{\bar{k}}$ as detailed in Eqs. (32a) to (32d). Generally, these expressions can be quite lengthy, so henceforth we confine them to Appendix B. Commutators depending on momenta which (as we shall shortly show) propagate in the final linear theory are denoted by $(\hat{\pi}!)$. These are significant as they are presumed to persist even when the full nonlinear Dirac–Bergmann algorithm is terminated, except possibly on strongly coupled spacetimes. Constant terms only arise in brackets between conjugate PiCs ($\boxed{\cdot}$), and then only if both PiCs have non-vanishing mass parameters. Since all the PiC mass parameters vanish in Case ^{*6}26, no constant terms can arise. The linearised theory is sensitive only to these constant terms, but we see from (33) that the conjugate PiCs also commute in the nonlinear Case ^{*6}26.

Let us now consider the consistency of the PiCs, and implement the Dirac–Bergmann algorithm for the linearised theory. Within the PiC shell, we encounter the following SiCs

$$\chi^\flat{}_{\perp\bar{k}} \approx -2\eta^{\flat\overline{ml}}\mathcal{D}^\flat{}_{\overline{m}}\hat{\hat{\pi}}^\flat{}_{\overline{kl}}, \tag{34a}$$
$$\hat{\hat{\chi}}^\flat{}_{\perp\overline{kl}} \approx 2\hat{\hat{\pi}}^\flat{}_{\overline{kl}} - \frac{1}{6}\epsilon^\flat{}_{\overline{klm}\perp}\eta^{\flat\overline{mn}}\mathcal{D}^\flat{}_{\overline{n}}{}^{\mathrm{P}}\hat{\pi}^\flat, \tag{34b}$$

where quantities linearised on the Minkowski background are denoted with ($\flat$). Also within this shell, we find $\mathcal{H}^\flat{}_\perp$ and $\mathcal{H}^\flat{}_{\perp\overline{k}}$ already vanish weakly, while the linear super-momentum and vector part of the rotational super-momentum give further sSFCs

$$\mathcal{H}^\flat{}_\alpha \approx -h^\flat{}_\alpha{}^{\overline{j}}\eta^{\flat\overline{kl}}\mathcal{D}^\flat{}_{\overline{k}}\hat{\hat{\pi}}^\flat{}_{\overline{jl}}, \tag{35a}$$

$$\mathcal{H}^\flat{}_{\overline{kl}} \approx 2\hat{\hat{\pi}}^\flat{}_{\overline{kl}} - \frac{1}{6}\epsilon^\flat{}_{\overline{klm}\perp}\eta^{\flat\overline{mn}}\mathcal{D}^\flat{}_{\overline{n}}{}^{\mathrm{P}}\hat{\pi}^\flat. \tag{35b}$$

The SiCs clearly vanish in the sSFC sub-shell, terminating the algorithm immediately. We find that $\mathcal{H}^\flat{}_\alpha$ is already implied by $\mathcal{H}^\flat{}_{\overline{kl}}$, which constitutes a total of three sSFCs. The PiCs are all FC, and the total number of iPFCs can be read off from (33). Recalling also the 10 sPFCs, and counting all FCs twice, we find that there is only one propagating D.o.F, as expected from Table II

$$\begin{aligned}1 = \frac{1}{2}\big(&80 - 2\times 10[\text{sPFC}] - 2\times 3[\text{sSFC}]\\ &- 2\times(1+3+5+1+3+3+5+5)[\text{iPFC}]\big).\end{aligned} \tag{36}$$

So, what is this D.o.F? We know that there are 26 undetermined multipliers, to match each of the iPFCs. Generically, this makes it very difficult to make sense of the equations of motion. However, we can make an educated guess by noticing that the functions $\hat{\hat{\varphi}}_{\overline{kl}}$ and ${}^{\mathrm{P}}\varphi$ are not PiCs and in the end, it turns out to be the 0^- tordion which is propagating. An application of (22) allows us to find the velocity of the pseudoscalar part of the torsion

$${}^{\mathrm{P}}\dot{\mathcal{T}}^\flat \approx -\frac{1}{2\hat{\alpha}_3}{}^{\mathrm{P}}\hat{\pi}^\flat - \frac{3}{4\hat{\beta}_3 m_{\mathrm{p}}{}^2}\epsilon^{\flat\overline{jkl}\perp}\mathcal{D}^\flat{}_{\overline{j}}\hat{\hat{\pi}}^\flat{}_{\overline{kl}}. \tag{37}$$

Conveniently, we see that this quantity makes no reference to undetermined multipliers in the final shell. Moreover, the same can be said of the acceleration

$${}^{\mathrm{P}}\ddot{\mathcal{T}}^\flat \approx \eta^{\flat\overline{jk}}\mathcal{D}^\flat{}_{\overline{j}}\mathcal{D}^\flat{}_{\overline{k}}{}^{\mathrm{P}}\mathcal{T}^\flat + \frac{4\hat{\beta}_3}{\hat{\alpha}_3}m_{\mathrm{p}}{}^2{}^{\mathrm{P}}\mathcal{T}^\flat, \tag{38}$$

which clearly describes a particle of mass

$$m \equiv 2\sqrt{\frac{|\hat{\beta}_3|}{|\hat{\alpha}_3|}}m_{\mathrm{p}}, \tag{39}$$

if $\hat{\beta}_3/\hat{\alpha}_3 > 0$. The unitarity conditions in Table II can now be decoded. The condition $\hat{\alpha}_3 < 0$ clearly wards off a 0^- ghost by inspection of (31), whereas $\hat{\beta}_3 < 0$ then prevents the 0^- from becoming tachyonic.

In the nonlinear theory, the PPM is no longer empty as shown in (33). We anticipate that the emergent commutators will ultimately result in a fundamentally different particle spectrum. Particlularly, we see from Eqs. (32c) and (32d) that $\varphi_{\perp\overline{k}}$, $\tilde{\varphi}_{\overline{kl}}$, $\hat{\varphi}_{\perp\overline{kl}}$ and ${}^{\mathrm{T}}\varphi_{\overline{klo}}$ are all demoted from iPFCs to iPSCs so long as 0^- is activated. Possibly, 0^- becomes strongly coupled on some privilaged surface, but since the converse is unlikely to be true we conclude that a linear gauge symmetry is *generally broken*.

To see one way in which this might affect the outcome, imagine that none of the sSFCs are degenerate in the full nonlinear theory, but that they still encode the iSFCs (which therefore need not appear in the final count). The nonlinear theory would then be expected to propagate two D.o.F

$$\begin{aligned}2 \overset{?}{=} \frac{1}{2}\big(&80 - 2\times 10[\text{sPFC}] - 2\times 10[\text{sSFC}]\\ &- 2\times(1+1+3+5)[\text{iPFC}]\\ &- (3+5+3+5)[\text{iPSC}]\big),\end{aligned} \tag{40}$$

suggesting that somehow one D.o.F from the 1^+ sector (i.e. the only J^P other than 0^- which is not primarily constrained), is generally activated, but becomes strongly coupled on Minkowski spacetime. It is not clear what this would look like, and we emphasise that the specific scenario in (40) is unlikely to be the one which is realised. The full picture can only be revealed by performing the nonlinear Dirac–Bergmann analysis, beginning from (33). Following treatments of simpler cases of PGT$^{\mathrm{q}+}$ in [9], we will not go this far. However, we think it likely that any activation of the 1^+ sector will damage the unitarity of the theory, since we see from (31) that $\hat{\hat{\pi}}_{\overline{kl}}\hat{\hat{\pi}}^{\overline{kl}}$ has a negative contribution to the energy, by the same condition $\hat{\beta}_3 < 0$ that upholds the unitarity of the 0^- mode. For further discussion of the 'positive energy test', we direct the reader to Appendix A.

Finally, (33) may also indicate that the nonlinear theory violates causality. We refer to the test set out in [8] and implemented in [9], whereby the PPM rank is required not to depend on the values of the fields and their momenta. This test is only nontrivial when the linearised PPM is populated by constant mass parameters: the nonlinear PPM pseudodeterminant must then be positive-definite within the final shell. Even this calculation is not necessary in the case at hand, since the linear PPM is empty and the nonlinear PPM is populated by momenta.

B. Case 28

Since Case 28 has fewer PiCs than Case *6 26, the kinetic part of the Hamiltonian is more extensive

$$\begin{aligned}\mathcal{H}_{\mathrm{T}} = \frac{b}{96}\Bigg(&\frac{6\big(\vec{\varphi}_{\overline{k}}\vec{\varphi}^{\overline{k}} + 2\hat{\varphi}_{\perp\overline{kl}}\hat{\varphi}^{\perp\overline{kl}}\big)}{\hat{\alpha}_5}\\ &+ \frac{18\hat{\varphi}_{\overline{kl}}\hat{\varphi}^{\overline{kl}}}{\hat{\beta}_3} - \frac{{}^{\mathrm{P}}\varphi^2}{\hat{\alpha}_3}\Bigg) + \dots,\end{aligned} \tag{41}$$

while the PPM has fewer dimensions

$$
\begin{array}{c|cccccc|c}
 & \varphi & \varphi_{\perp\overline{k}} & \widetilde{\varphi}_{\overline{kl}} & \varphi_{\perp} & \widetilde{\varphi}_{\perp\overline{kl}} & {}^{\mathrm{T}}\varphi_{\overline{klo}} & \\
\hline
\varphi & \cdot & \cdot & \cdot & \boxed{\cdot} & \cdot & \cdot & 1 \\
\varphi_{\perp\overline{k}} & \cdot & \hat{\pi} & \cdot & \hat{\pi} & \hat{\pi} & \hat{\pi} & 3 \\
\widetilde{\varphi}_{\overline{kl}} & \cdot & \cdot & \hat{\pi} & \cdot & \boxed{\hat{\pi}} & \hat{\pi}! & 5 \\
\varphi_{\perp} & \boxed{\cdot} & \hat{\pi} & \cdot & \cdot & \cdot & \cdot & 1 \\
\widetilde{\varphi}_{\perp\overline{kl}} & \cdot & \hat{\pi} & \boxed{\hat{\pi}} & \cdot & \cdot & \cdot & 5 \\
{}^{\mathrm{T}}\varphi_{\overline{klo}} & \cdot & \hat{\pi} & \hat{\pi}! & \cdot & \cdot & \cdot & 5 \\
\hline
 & 1 & 3 & 5 & 1 & 5 & 5 &
\end{array}
\tag{42}
$$

Within the PiC shell, we find that $\varphi^\flat$ and $\widetilde{\varphi}^\flat{}_{\overline{kl}}$ already weakly vanish, leaving the following SiCs

$$\chi^\flat{}_{\perp\overline{k}} \approx -2\eta^{\flat\overline{ml}}\mathcal{D}^\flat{}_{\overline{m}}\hat{\hat{\pi}}^\flat{}_{\overline{kl}}, \tag{43a}$$

$$\chi^\flat{}_{\perp} \approx -\eta^{\flat\overline{ml}}\mathcal{D}^\flat{}_{\overline{m}}\vec{\hat{\pi}}^\flat{}_{\overline{l}}, \tag{43b}$$

$$\widetilde{\chi}^\flat{}_{\perp\overline{kl}} \approx \frac{1}{2}\mathcal{D}^\flat{}_{\langle\overline{k}}\vec{\hat{\pi}}^\flat{}_{\overline{l}\rangle}, \tag{43c}$$

$$
\begin{aligned}
{}^{\mathrm{T}}\chi^\flat{}_{\overline{klo}} \approx{}& \frac{1}{2}\mathcal{D}^\flat{}_{\overline{o}}\hat{\hat{\pi}}^\flat{}_{\perp\overline{kl}} + \frac{1}{2}\mathcal{D}^\flat{}_{[\overline{l}}\hat{\hat{\pi}}^\flat{}_{\perp\overline{k}]\overline{o}} \\
&+ \frac{3}{4}\eta^\flat{}_{\overline{o}[\overline{k}|}\eta^{\flat\overline{ij}}\mathcal{D}^\flat{}_{\overline{i}}\hat{\hat{\pi}}^\flat{}_{\perp|\overline{l}]\overline{j}},
\end{aligned}
\tag{43d}
$$

which do not give rise to any TiCs. Also within the PiC shell, the following sSFCs appear

$$\mathcal{H}^\flat{}_{\alpha} \approx -h^\flat{}_{\alpha}{}^{\overline{j}}\eta^{\flat\overline{kl}}\mathcal{D}^\flat{}_{\overline{k}}\hat{\hat{\pi}}^\flat{}_{\overline{jl}}, \tag{44a}$$

$$\mathcal{H}^\flat{}_{\overline{kl}} \approx 2\hat{\hat{\pi}}^\flat{}_{\overline{kl}} - \frac{1}{6}\epsilon^\flat{}_{\overline{klm}\perp}\eta^{\flat\overline{mn}}\mathcal{D}^\flat{}_{\overline{n}}{}^{\mathrm{P}}\hat{\pi}^\flat + \mathcal{D}^\flat{}_{[\overline{k}}\vec{\hat{\pi}}^\flat{}_{\overline{l}]}, \tag{44b}$$

$$\mathcal{H}^\flat{}_{\perp\overline{k}} \approx \eta^{\flat\overline{jl}}\mathcal{D}^\flat{}_{\overline{j}}\hat{\hat{\pi}}^\flat{}_{\perp\overline{kl}}. \tag{44c}$$

In this case it is easiest to restrict to sub-shells using the SiCs and sSFCs simultaneously. We first note that $\mathcal{H}^\flat{}_{\perp\overline{k}}$ restricts $\hat{\hat{\pi}}^\flat{}_{\perp\overline{kl}}$ to be solenoidal, dual to the gradient of a scalar, and thus eliminates two D.o.F. The remaining D.o.F is eliminated by ${}^{\mathrm{T}}\chi^\flat{}_{\overline{klo}}$. Similarly, $\chi^\flat{}_{\perp}$ restricts $\vec{\hat{\pi}}^\flat{}_{\overline{k}}$ to a solenoidal axial vector, removing one D.o.F. A further D.o.F is removed by substituting $\mathcal{H}^\flat{}_{\overline{kl}}$ into $\mathcal{H}^\flat{}_{\alpha}$, and a final D.o.F is removed by $\widetilde{\chi}^\flat{}_{\perp\overline{kl}}$. Separately, $\mathcal{H}^\flat{}_{\overline{kl}}$ removes three D.o.Fs. All the PiCs and SiCs are FC, and one D.o.F remains, as expected from Table II

$$
\begin{aligned}
1 = \frac{1}{2}\big(&80 - 2\times 10[\text{sPFC}] - 2\times(1+3+2)[\text{sSFC}] \\
&- 2\times(1+3+5+1+5+5)[\text{iPFC}] \\
&- 2\times(1+1+1)[\text{iSFC}]\big).
\end{aligned}
\tag{45}
$$

C. Case ^{*5}25

The Hamiltonian takes the form

$$
\begin{aligned}
\mathcal{H}_{\mathrm{T}} = \frac{b}{96}\Bigg(&\frac{4\big(\varphi^2 + 9\varphi_{\perp\overline{k}}\varphi^{\perp\overline{k}}\big)}{\hat{\beta}_2} \\
&+ \frac{18\hat{\varphi}_{\overline{kl}}\hat{\varphi}^{\overline{kl}}}{\hat{\beta}_3} - \frac{{}^{\mathrm{P}}\varphi^2}{\hat{\alpha}_3}\Bigg) + \ldots,
\end{aligned}
\tag{46}
$$

The PPM is:

$$
\begin{array}{c|cccccc|c}
 & \widetilde{\varphi}_{\overline{kl}} & \varphi_{\perp} & \hat{\varphi}_{\perp\overline{kl}} & \vec{\varphi}_{\overline{k}} & \widetilde{\varphi}_{\perp\overline{kl}} & {}^{\mathrm{T}}\varphi_{\overline{klo}} & \\
\hline
\widetilde{\varphi}_{\overline{kl}} & \hat{\pi} & \cdot & \cdot & \cdot & \boxed{\cdot} & \hat{\pi}! & 5 \\
\varphi_{\perp} & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & 1 \\
\hat{\varphi}_{\perp\overline{kl}} & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & 3 \\
\vec{\varphi}_{\overline{k}} & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & 3 \\
\widetilde{\varphi}_{\perp\overline{kl}} & \boxed{\cdot} & \cdot & \cdot & \cdot & \cdot & \cdot & 5 \\
{}^{\mathrm{T}}\varphi_{\overline{klo}} & \hat{\pi}! & \cdot & \cdot & \cdot & \cdot & \cdot & 5 \\
\hline
 & 5 & 1 & 3 & 3 & 5 & 5 &
\end{array}
\tag{47}
$$

Within the PiC shell, we have

$$\widetilde{\chi}^\flat{}_{\overline{kl}} \approx -\mathcal{D}^\flat{}_{\langle\overline{k}}\hat{\pi}^\flat{}_{\perp\overline{l}\rangle} \tag{48a}$$

$$\chi^\flat{}_{\perp} \approx \hat{\pi}^\flat, \tag{48b}$$

$$\hat{\chi}^\flat{}_{\perp\overline{kl}} \approx 2\hat{\hat{\pi}}^\flat{}_{\overline{kl}} - \frac{1}{6}\epsilon^\flat{}_{\overline{klm}\perp}\eta^{\flat\overline{mn}}\mathcal{D}^\flat{}_{\overline{n}}{}^{\mathrm{P}}\hat{\pi}^\flat, \tag{48c}$$

$$\vec{\chi}^\flat{}_{\overline{k}} \approx 2\hat{\pi}^\flat{}_{\perp\overline{k}}, \tag{48d}$$

and this time, all 10 sSFCs persist in the PiC shell

$$\mathcal{H}^\flat{}_{\perp} \approx -\eta^{\flat\overline{kl}}\mathcal{D}^\flat{}_{\overline{k}}\hat{\pi}^\flat{}_{\perp\overline{l}} \tag{49a}$$

$$\mathcal{H}^\flat{}_{\alpha} \approx -\frac{1}{3}h^\flat{}_{\alpha}{}^{\overline{k}}\mathcal{D}^\flat{}_{\overline{k}}\hat{\pi}^\flat - h^\flat{}_{\alpha}{}^{\overline{k}}\eta^{\flat\overline{jl}}\mathcal{D}^\flat{}_{\overline{j}}\hat{\hat{\pi}}^\flat{}_{\overline{kl}}, \tag{49b}$$

$$\mathcal{H}^\flat{}_{\overline{kl}} \approx 2\hat{\hat{\pi}}^\flat{}_{\overline{kl}} - \frac{1}{6}\epsilon^\flat{}_{\overline{klm}\perp}\eta^{\flat\overline{mn}}\mathcal{D}^\flat{}_{\overline{n}}{}^{\mathrm{P}}\hat{\pi}^\flat, \tag{49c}$$

$$\mathcal{H}^\flat{}_{\perp\overline{k}} \approx \hat{\pi}^\flat{}_{\perp\overline{k}}. \tag{49d}$$

We find that $\mathcal{H}^\flat{}_{\perp\overline{k}}$ and $\mathcal{H}^\flat{}_{\overline{kl}}$ each remove three D.o.F, while $\chi^\flat{}_{\perp}$ removes one D.o.F; the remaining sSFCs and SiCs are then implied, and the PiCs and SiCs are FC. Once again, one D.o.F remains as expected from Table II

$$
\begin{aligned}
1 = \frac{1}{2}\big(&80 - 2\times 10[\text{sPFC}] - 2\times(3+3)[\text{sSFC}] \\
&- 2\times(5+1+3+3+5+5)[\text{iPFC}] \\
&- 2\times 1[\text{iSFC}]\big).
\end{aligned}
\tag{50}
$$

D. Case 24

$$
\begin{aligned}
\mathcal{H}_{\mathrm{T}} = \frac{b}{96}\Bigg(&\frac{6\big(\vec{\varphi}_{\overline{k}}\vec{\varphi}^{\overline{k}2} + 2\hat{\varphi}_{\perp\overline{kl}}\hat{\varphi}^{\perp\overline{kl}}\big)}{\hat{\alpha}_5} \\
&+ \frac{4\big(\varphi^2 + 9\varphi_{\perp\overline{k}}\varphi^{\perp\overline{k}}\big)}{\hat{\beta}_2} + \frac{18\hat{\varphi}_{\overline{kl}}\hat{\varphi}^{\overline{kl}}}{\hat{\beta}_3} - \frac{{}^{\mathrm{P}}\varphi^2}{\hat{\alpha}_3}\Bigg) + \ldots,
\end{aligned}
\tag{51}
$$

The PPM is:

$$
\begin{array}{c|cccc|c}
 & \widetilde{\varphi}_{\overline{kl}} & \varphi_{\perp} & \widetilde{\varphi}_{\perp\overline{kl}} & {}^{\mathrm{T}}\varphi_{\overline{klo}} & \\
\hline
\widetilde{\varphi}_{\overline{kl}} & \hat{\pi} & \cdot & \boxed{\hat{\pi}} & \hat{\pi}! & 5 \\
\varphi_{\perp} & \cdot & \cdot & \cdot & \cdot & 1 \\
\widetilde{\varphi}_{\perp\overline{kl}} & \boxed{\hat{\pi}} & \cdot & \cdot & \cdot & 5 \\
{}^{\mathrm{T}}\varphi_{\overline{klo}} & \hat{\pi}! & \cdot & \cdot & \cdot & 5 \\
\hline
 & 5 & 1 & 5 & 5 &
\end{array}
\tag{52}
$$

Within the PiC shell, we have the following SiCs

$$\widetilde{\chi}^\flat{}_{\overline{kl}} \approx -\mathcal{D}^\flat{}_{\langle\overline{k}}\hat{\pi}^\flat{}_{\perp\overline{l}\rangle} \tag{53a}$$

$$\chi^\flat{}_\perp \approx \hat{\pi}^\flat - \eta^{\flat\overline{kl}}\mathcal{D}^\flat{}_{\overline{k}}\overrightarrow{\hat{\pi}}^\flat{}_{\overline{l}}, \tag{53b}$$

$$\widetilde{\chi}^\flat{}_{\perp\overline{kl}} \approx \frac{1}{2}\mathcal{D}^\flat{}_{\langle\overline{k}}\overrightarrow{\hat{\pi}}^\flat{}_{\overline{l}\rangle}, \tag{53c}$$

$$\begin{aligned}{}^{\mathrm{T}}\chi^\flat{}_{\overline{klo}} \approx{}& \frac{1}{2}\mathcal{D}^\flat{}_{\overline{o}}\hat{\hat{\pi}}^\flat{}_{\perp\overline{kl}} + \frac{1}{2}\mathcal{D}^\flat{}_{[\overline{l}}\hat{\hat{\pi}}^\flat{}_{\perp\overline{k}]\overline{o}} \\ &+ \frac{3}{4}\eta^\flat{}_{\overline{o}[\overline{k}|}\eta^{\flat\overline{ij}}\mathcal{D}^\flat{}_{\overline{i}}\hat{\hat{\pi}}^\flat{}_{\perp|\overline{l}|\overline{j}},\end{aligned} \tag{53d}$$

and the following sPFCs

$$\mathcal{H}^\flat{}_\perp \approx -\eta^{\flat\overline{kl}}\mathcal{D}^\flat{}_{\overline{k}}\hat{\pi}^\flat{}_{\perp\overline{l}} \tag{54a}$$

$$\mathcal{H}^\flat{}_\alpha \approx -\frac{1}{3}h^\flat{}_\alpha{}^{\overline{k}}\mathcal{D}^\flat{}_{\overline{k}}\hat{\pi}^\flat - h^\flat{}_\alpha{}^{\overline{k}}\eta^{\flat\overline{jl}}\mathcal{D}^\flat{}_{\overline{j}}\hat{\hat{\pi}}^\flat{}_{\overline{kl}}, \tag{54b}$$

$$\mathcal{H}^\flat{}_{\overline{kl}} \approx 2\hat{\hat{\pi}}^\flat{}_{\overline{kl}} - \frac{1}{6}\epsilon^\flat{}_{\overline{klm}\perp}\eta^{\flat\overline{mn}}\mathcal{D}^\flat{}_{\overline{n}}{}^{\mathrm{P}}\hat{\pi}^\flat + \mathcal{D}^\flat{}_{[\overline{k}}\overrightarrow{\hat{\pi}}^\flat{}_{\overline{l}]}, \tag{54c}$$

$$\mathcal{H}^\flat{}_{\perp\overline{k}} \approx \hat{\pi}^\flat{}_{\perp\overline{k}} + \eta^{\flat\overline{jl}}\mathcal{D}^\flat{}_{\overline{j}}\hat{\hat{\pi}}^\flat{}_{\perp\overline{kl}}. \tag{54d}$$

It is clear from the PiC shell that Case 24 has much in common with Case 28, and again we will implement the SiCs and sSFCs simultaneously. Firstly, we find that ${}^{\mathrm{T}}\chi^\flat{}_{\overline{klo}}$ constitutes an overdetermined system in $\hat{\hat{\pi}}^\flat{}_{\perp\overline{kl}}$, which vanishes and takes with it three D.o.F. Consequently, from $\mathcal{H}^\flat{}_{\perp\overline{k}}$, we see that $\hat{\pi}^\flat{}_{\perp\overline{k}}$ must vanish along with another three D.o.F, such that $\mathcal{H}^\flat{}_\perp$ and $\widetilde{\chi}^\flat{}_{\overline{kl}}$ vanish automatically. Similarly, $\widetilde{\chi}^\flat{}_{\perp\overline{kl}}$ is an overdetermined system in $\hat{\hat{\pi}}^\flat{}_{\perp\overline{kl}}$, which vanishes with another three D.o.F; $\chi^\flat{}_\perp$ then causes $\hat{\pi}^\flat$ to vanish with one D.o.F. As before, one D.o.F propagates in accordance with Table II

$$\begin{aligned}1 = \frac{1}{2}\big(&80 - 2\times 10[\text{sPFC}] - 2\times(3+3)[\text{sSFC}] \\ &- 2\times(5+1+5+5)[\text{iPFC}] \\ &- 2\times(1+3+3)[\text{iSFC}]\big).\end{aligned} \tag{55}$$

E. Case 32

$$\begin{aligned}\mathcal{H}_{\mathrm{T}} = \frac{b}{96}\Bigg(&\frac{6\big(3\varphi_{\perp\overline{k}}\varphi^{\perp\overline{k}2} + 2\widetilde{\varphi}_{\overline{kl}}\widetilde{\varphi}^{\overline{kl}}\big)}{\hat{\beta}_1} \\ &+ \frac{36\hat{\hat{\varphi}}_{\overline{kl}}\hat{\hat{\varphi}}^{\overline{kl}}}{\hat{\beta}_1 + 2\hat{\beta}_3} - \frac{{}^{\mathrm{P}}\varphi^2}{\hat{\alpha}_3}\Bigg) + \ldots,\end{aligned} \tag{56}$$

For the first time, we encounter non-vanishing mass parameters in the PiCs, specifically in $\hat{\hat{\varphi}}^\flat{}_{\perp\overline{kl}}$ and ${}^{\mathrm{T}}\varphi^\flat{}_{\overline{klo}}$. We anticipate the nonvanishing commutators $\left\{\hat{\hat{\varphi}}^\flat{}_{\perp\overline{kl}}, \hat{\hat{\chi}}^\flat{}_{\perp\overline{ij}}\right\} \approx \mathcal{O}(1)$ – noting that the natural conjugate $\hat{\hat{\varphi}}^\flat{}_{\overline{kl}}$ is not a PiC – and $\left\{{}^{\mathrm{T}}\varphi^\flat{}_{\overline{klo}}, {}^{\mathrm{T}}\chi^\flat{}_{\overline{ijn}}\right\} \approx \mathcal{O}(1)$. These PiCs and SiCs will be SC, allowing for the determination of their multipliers. The PPM of this theory is remarkable, since it remains empty even in the nonlinear regime:

$$\begin{array}{r|cccccc|c}
 & & & \downarrow & & & \downarrow & \\
 & \varphi & \varphi_\perp & \hat{\hat{\varphi}}_{\perp\overline{kl}} & \overrightarrow{\varphi}_{\overline{k}} & \widetilde{\varphi}_{\perp\overline{kl}} & {}^{\mathrm{T}}\varphi_{\overline{klo}} & \\
\varphi & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & 1 \\
\varphi_\perp & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & 1 \\
\rightarrow\ \hat{\hat{\varphi}}_{\perp\overline{kl}} & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & 3 \\
\overrightarrow{\varphi}_{\overline{k}} & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & 3 \\
\widetilde{\varphi}_{\perp\overline{kl}} & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & 5 \\
\rightarrow\ {}^{\mathrm{T}}\varphi_{\overline{klo}} & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & 5 \\
\hline
 & 1 & 1 & 3 & 3 & 5 & 5 &
\end{array} \tag{57}$$

Within the PiC shell, we find the following SiCs

$$\chi^\flat \approx -\eta^{\flat\overline{kl}}\mathcal{D}^\flat{}_{\overline{k}}\hat{\pi}^\flat{}_{\perp\overline{l}}, \tag{58a}$$

$$\begin{aligned}\hat{\hat{\chi}}^\flat{}_{\perp\overline{kl}} \approx{}& -\frac{\hat{\beta}_1 + 2\hat{\beta}_3}{\hat{\beta}_1 - \hat{\beta}_3}\hat{\hat{\pi}}^\flat{}_{\overline{kl}} \\ &- \frac{1}{6}\epsilon^\flat{}_{\overline{klm}\perp}\eta^{\flat\overline{mn}}\mathcal{D}^\flat{}_{\overline{n}}{}^{\mathrm{P}}\hat{\pi}^\flat \\ &+ \frac{9\hat{\beta}_1\hat{\beta}_3}{(\hat{\beta}_1 - \hat{\beta}_3)(\hat{\beta}_1 + 2\hat{\beta}_3)}\hat{\hat{\varphi}}^\flat{}_{\overline{kl}},\end{aligned} \tag{58b}$$

$$\overrightarrow{\chi}^\flat{}_{\overline{k}} \approx -\hat{\pi}^\flat{}_{\perp\overline{k}}, \tag{58c}$$

$$\widetilde{\chi}^\flat{}_{\perp\overline{kl}} \approx \widetilde{\hat{\pi}}^\flat{}_{\overline{kl}}, \tag{58d}$$

$${}^{\mathrm{T}}\chi^\flat{}_{\overline{klo}} \approx 4\hat{\beta}_1 m_{\mathrm{p}}{}^2\,{}^{\mathrm{T}}\mathcal{T}^\flat{}_{\overline{klo}}\,. \tag{58e}$$

Note the appearance of field strengths, specifically the torsion in $\hat{\hat{\varphi}}^\flat{}_{\overline{kl}}$ and ${}^{\mathrm{T}}\mathcal{T}^\flat{}_{\overline{klo}}$. Whilst these somewhat complicate the analysis, they naturally appear with the masses. We also mark the first apparent instance of a TiC accompanying $\widetilde{\chi}^\flat{}_{\perp\overline{kl}}$. Using the notation $\zeta = \dot{\chi}$, this may be written as

$$\widetilde{\zeta}^\flat{}_{\perp\overline{kl}} \approx \frac{4}{3}\eta^{\flat\overline{ij}}\mathcal{D}^\flat{}_{\overline{i}}\,{}^{\mathrm{T}}\chi^\flat{}_{\langle\overline{k}|\overline{j}|\overline{l}\rangle}, \tag{59}$$

which then vanishes in the SiC shell. The PiC shell contains the following sSFCs:

$$\mathcal{H}^\flat{}_\perp \approx -\eta^{\flat\overline{kl}}\mathcal{D}^\flat{}_{\overline{k}}\hat{\pi}^\flat{}_{\perp\overline{l}} \tag{60a}$$

$$\mathcal{H}^\flat{}_\alpha \approx -h^\flat{}_\alpha{}^{\overline{k}}\eta^{\flat\overline{jl}}\mathcal{D}^\flat{}_{\overline{j}}\hat{\hat{\pi}}^\flat{}_{\overline{kl}} - h^\flat{}_\alpha{}^{\overline{k}}\eta^{\flat\overline{jl}}\mathcal{D}^\flat{}_{\overline{j}}\widetilde{\hat{\pi}}^\flat{}_{\overline{kl}}, \tag{60b}$$

$$\mathcal{H}^\flat{}_{\overline{kl}} \approx 2\hat{\hat{\pi}}^\flat{}_{\overline{kl}} - \frac{1}{6}\epsilon^\flat{}_{\overline{klm}\perp}\eta^{\flat\overline{mn}}\mathcal{D}^\flat{}_{\overline{n}}{}^{\mathrm{P}}\hat{\pi}^\flat, \tag{60c}$$

$$\mathcal{H}^\flat{}_{\perp\overline{k}} \approx \hat{\pi}^\flat{}_{\perp\overline{k}}. \tag{60d}$$

Since two of the PiC chains are known to be self-terminating, the algorithm concludes quite quickly. As with Case *5 25, $\mathcal{H}^\flat{}_{\perp\overline{k}}$ and $\mathcal{H}^\flat{}_{\overline{kl}}$ each eliminate three D.o.F. Another five D.o.F are then removed by $\widetilde{\chi}^\flat{}_{\perp\overline{kl}}$, with the remaining SiCs and sSFCs automatically satisfied. The one remaining D.o.F is again expected from Table II

$$\begin{aligned}1 = \frac{1}{2}\big(&80 - 2\times 10[\text{sPFC}] - 2\times(3+3)[\text{sSFC}] \\ &- 2\times(1+1+3+5)[\text{iPFC}] - (3+5)[\text{iPSC}] \\ &- 2\times 5[\text{iSFC}] - (3+5)[\text{iSSC}]\big).\end{aligned} \tag{61}$$

F. Case 20

$$\mathcal{H}_{\mathrm{T}} = \frac{b}{96}\Bigg(\frac{4\varphi^2}{\hat{\beta}_2} + \frac{12\widetilde{\varphi}_{\overline{kl}}\widetilde{\varphi}^{\overline{kl}}}{\hat{\beta}_1} + \frac{36\hat{\varphi}_{\overline{kl}}\hat{\varphi}^{\overline{kl}}}{\hat{\beta}_1+2\hat{\beta}_3} + \frac{36\varphi_{\perp\overline{k}}\varphi^{\perp\overline{k}}}{2\hat{\beta}_1+\hat{\beta}_2} - \frac{{}^{\mathrm{P}}\varphi^2}{\hat{\alpha}_3}\Bigg) + \dots, \tag{62}$$

The analysis of Case 20 is quite similar to Case 32. Mass parameters again accompany the PiCs, and we expect $\hat{\varphi}^\flat{}_{\perp\overline{kl}}$, $\vec{\varphi}^\flat{}_{\overline{k}}$ and ${}^{\mathrm{T}}\varphi^\flat{}_{\overline{klo}}$ to not commute with their respective SiCs on the final shell. Once again, the PPM is empty both before and after linearisation:

$$\begin{array}{cc|ccccc|c}
 & & & \downarrow & \downarrow & & \downarrow & \\
 & & \varphi_\perp & \hat{\varphi}_{\perp\overline{kl}} & \vec{\varphi}_{\overline{k}} & \widetilde{\varphi}_{\perp\overline{kl}} & {}^{\mathrm{T}}\varphi_{\overline{klo}} & \\
\hline
 & \varphi_\perp & \cdot & \cdot & \cdot & \cdot & \cdot & 1 \\
\rightarrow & \hat{\varphi}_{\perp\overline{kl}} & \cdot & \cdot & \cdot & \cdot & \cdot & 3 \\
\rightarrow & \vec{\varphi}_{\overline{k}} & \cdot & \cdot & \cdot & \cdot & \cdot & 3 \\
 & \widetilde{\varphi}_{\perp\overline{kl}} & \cdot & \cdot & \cdot & \cdot & \cdot & 5 \\
\rightarrow & {}^{\mathrm{T}}\varphi_{\overline{klo}} & \cdot & \cdot & \cdot & \cdot & \cdot & 5 \\
\hline
 & & 1 & 3 & 3 & 5 & 5 &
\end{array} \tag{63}$$

Within the PiC shell, we first the following SiCs

$$\chi^\flat{}_\perp \approx \hat{\pi}^\flat, \tag{64a}$$

$$\hat{\chi}^\flat{}_{\perp\overline{kl}} \approx -\frac{\hat{\beta}_1+2\hat{\beta}_3}{\hat{\beta}_1-\hat{\beta}_3}\hat{\pi}^\flat{}_{\overline{kl}} - \frac{1}{6}\epsilon^\flat{}_{\overline{klm}\perp}\eta^{\flat\overline{mn}}\mathcal{D}^\flat{}_{\overline{n}}{}^{\mathrm{P}}\hat{\pi}^\flat + \frac{9\hat{\beta}_1\hat{\beta}_3}{(\hat{\beta}_1-\hat{\beta}_3)(\hat{\beta}_1+2\hat{\beta}_3)}\hat{\varphi}^\flat{}_{\overline{kl}}, \tag{64b}$$

$$\vec{\chi}^\flat{}_{\overline{k}} \approx -\frac{\hat{\beta}_1+2\hat{\beta}_2}{\hat{\beta}_1-\hat{\beta}_2}\hat{\pi}^\flat{}_{\perp\overline{k}} \tag{64c}$$

$$+ \frac{9\hat{\beta}_1\hat{\beta}_2}{(\hat{\beta}_1-\hat{\beta}_2)(2\hat{\beta}_1+\hat{\beta}_2)}\varphi^\flat{}_{\perp\overline{k}}, \tag{64d}$$

$$\widetilde{\chi}^\flat{}_{\perp\overline{kl}} \approx \widetilde{\hat{\pi}}^\flat{}_{\overline{kl}}, \tag{64e}$$

$${}^{\mathrm{T}}\chi^\flat{}_{\overline{klo}} \approx 4\hat{\beta}_1 m_{\mathrm{p}}{}^2\,{}^{\mathrm{T}}\mathcal{T}^\flat{}_{\overline{klo}}\,. \tag{64f}$$

This time, two TiCs appear, but uppon rearranging both may eventually be written in terms of the iSSCs, and are satisfied automatically

$$\zeta^\flat{}_\perp \approx \eta^{\flat\overline{ij}}\mathcal{D}^\flat{}_{\overline{i}}\vec{\chi}^\flat{}_{\overline{j}}, \tag{65a}$$

$$\widetilde{\zeta}^\flat{}_{\perp\overline{kl}} \approx \frac{4}{3}\eta^{\flat\overline{ij}}\mathcal{D}^\flat{}_{\overline{i}}{}^{\mathrm{T}}\chi^\flat{}_{\langle\overline{k}|\overline{j}|\overline{l}\rangle} - \frac{1}{2}\mathcal{D}^\flat{}_{\langle\overline{k}}\vec{\chi}^\flat{}_{\overline{l}\rangle}. \tag{65b}$$

The sSFC content in the PiC shell is largely the same as that of Case 32, with the only difference marked in the linear super-momentum

$$\mathcal{H}^\flat{}_\alpha \approx -\frac{1}{3}h^\flat{}_\alpha{}^{\overline{k}}\mathcal{D}^\flat{}_{\overline{k}}\hat{\pi}^\flat - h^\flat{}_\alpha{}^{\overline{k}}\eta^{\flat\overline{jl}}\mathcal{D}^\flat{}_{\overline{j}}\hat{\hat{\pi}}^\flat{}_{\overline{kl}} - h^\flat{}_\alpha{}^{\overline{k}}\eta^{\flat\overline{jl}}\mathcal{D}^\flat{}_{\overline{j}}\widetilde{\hat{\pi}}^\flat{}_{\overline{kl}}. \tag{66a}$$

Aided by the additional conjugate pair of constraints, the algorithm terminates even faster than with Case 32: we see that one and five D.o.F are removed by each of $\chi^\flat{}_\perp$ and $\widetilde{\varphi}^\flat{}_{\perp\overline{kl}}$. As before, the one propagating D.o.F is confirmed from Table II

$$\begin{aligned}1 = \frac{1}{2}\big(&80 - 2\times 10[\mathrm{sPFC}] - 2\times(3+3)[\mathrm{sSFC}] \\ &- 2\times(1+5)[\mathrm{iPFC}] - (3+3+5)[\mathrm{iPSC}] \\ &- 2\times(1+5)[\mathrm{iSFC}] - (3+3+5)[\mathrm{iSSC}]\big).\end{aligned} \tag{67}$$

IV. MASSLESS THEORIES

A. Case 17

$$\mathcal{H}_{\mathrm{T}} = \frac{b}{32}\Bigg(\frac{2(\vec{\varphi}_{\overline{k}}\vec{\varphi}^{\overline{k}} + 2\hat{\varphi}_{\perp\overline{kl}}\hat{\varphi}^{\perp\overline{kl}})}{\hat{\alpha}_5} - \frac{3\varphi_{\perp\overline{k}}\varphi^{\perp\overline{k}} + 2\widetilde{\varphi}_{\overline{kl}}\widetilde{\varphi}^{\overline{kl}}}{\hat{\beta}_3}\Bigg) + \dots, \tag{68}$$

As mentioned in Section II C, the evaluation of the PPM is complicated by the appearance of torsion in PiCs belonging to the translational sector, which generates derivatives of the Dirac function. Fortunately, these derivatives may be discarded up to a surface term even within the PiC shell, and the entire matrix may be written in terms of parallel momenta as before

$$\begin{array}{cc|cccccc|c}
 & & & \downarrow & & \downarrow & & \downarrow & \\
 & & \varphi & \hat{\varphi}_{\overline{kl}} & \varphi_\perp & {}^{\mathrm{P}}\varphi & \widetilde{\varphi}_{\perp\overline{kl}} & {}^{\mathrm{T}}\varphi_{\overline{klo}} & \\
\hline
 & \varphi & \cdot & \hat{\pi} & \cdot & \cdot & \cdot & \cdot & 1 \\
\rightarrow & \hat{\varphi}_{\overline{kl}} & \hat{\pi} & \cdot & \hat{\pi} & \hat{\pi} & \hat{\pi} & \hat{\pi} & 3 \\
 & \varphi_\perp & \cdot & \hat{\pi} & \cdot & \cdot & \cdot & \cdot & 1 \\
\rightarrow & {}^{\mathrm{P}}\varphi & \cdot & \hat{\pi} & \cdot & \cdot & \cdot & \cdot & 1 \\
 & \widetilde{\varphi}_{\perp\overline{kl}} & \cdot & \hat{\pi} & \cdot & \cdot & \cdot & \cdot & 5 \\
\rightarrow & {}^{\mathrm{T}}\varphi_{\overline{klo}} & \cdot & \hat{\pi} & \cdot & \cdot & \cdot & \cdot & 5 \\
\hline
 & & 1 & 3 & 1 & 1 & 5 & 5 &
\end{array} \tag{69}$$

Due to the appearance of mass parameters, we will expect $\hat{\varphi}^\flat{}_{\overline{kl}}$, ${}^{\mathrm{P}}\varphi^\flat$ and ${}^{\mathrm{T}}\varphi^\flat{}_{\overline{klo}}$ not to commute with their SiCs in the final shell. Within the PiC shell, we find the following

$$\chi^\flat \approx -\eta^{\flat\overline{kl}}\mathcal{D}^\flat{}_{\overline{k}}\hat{\pi}^\flat{}_{\perp\bar{l}}, \tag{70a}$$

$$\hat{\chi}^\flat{}_{\overline{kl}} \approx -\frac{2\hat{\beta}_3}{\hat{\alpha}_5}m_{\mathrm{p}}{}^2\hat{\hat{\pi}}^\flat{}_{\perp\overline{kl}} - \mathcal{D}^\flat{}_{[\overline{k}}\hat{\pi}^\flat{}_{\perp\bar{l}]} - 8\hat{\beta}_3 m_{\mathrm{p}}{}^2\mathcal{D}^\flat{}_{[\overline{k}}\vec{\mathcal{T}}^\flat{}_{\bar{l}]}, \tag{70b}$$

$$\chi^\flat{}_\perp \approx -\eta^{\flat\overline{kl}}\mathcal{D}^\flat{}_{\overline{k}}\vec{\hat{\pi}}^\flat{}_{\bar{l}}, \tag{70c}$$

$$^{\mathrm{P}}\chi^\flat \approx 2\epsilon^{\flat\overline{\bar{j}kl}\perp}\mathcal{D}^\flat{}_{\bar{j}}\hat{\hat{\pi}}^\flat{}_{\perp\overline{kl}} - 8\hat{\beta}_3 m_{\mathrm{p}}{}^2\,{}^{\mathrm{P}}\mathcal{T}^\flat, \tag{70d}$$

$$\widetilde{\chi}^\flat{}_{\perp\overline{kl}} \approx \widetilde{\hat{\pi}}^\flat{}_{\overline{kl}} + \frac{1}{2}\mathcal{D}^\flat{}_{\langle\overline{k}}\vec{\hat{\pi}}^\flat{}_{\bar{l}\rangle}, \tag{70e}$$

$$^{\mathrm{T}}\chi^\flat{}_{\overline{klo}} \approx \frac{1}{2}\mathcal{D}^\flat{}_{\overline{o}}\hat{\hat{\pi}}^\flat{}_{\perp\overline{kl}} + \frac{1}{2}\mathcal{D}^\flat{}_{[\bar{l}}\hat{\hat{\pi}}^\flat{}_{\perp\bar{k}]\overline{o}} + \frac{3}{4}\eta^\flat{}_{\overline{o}[\bar{k}|}\eta^{\flat\overline{\bar{i}\bar{j}}}\mathcal{D}^\flat{}_{\bar{i}}\hat{\hat{\pi}}^\flat{}_{\perp|\bar{l}]\bar{j}} - 8\hat{\beta}_3 m_{\mathrm{p}}{}^2\,{}^{\mathrm{T}}\mathcal{T}^\flat{}_{\overline{klo}}\,. \tag{70f}$$

Among these, we note that a TiC accompanies $\widetilde{\chi}^\flat{}_{\perp\overline{kl}}$, but may be expressed in terms of $^{\mathrm{T}}\chi^\flat{}_{\overline{klo}}$ by precisely (59). Within the PiC shell, the sSFCs are

$$\mathcal{H}^\flat{}_\perp \approx -\eta^{\flat\overline{kl}}\mathcal{D}^\flat{}_{\overline{k}}\hat{\pi}^\flat{}_{\perp\bar{l}}, \tag{71a}$$

$$\mathcal{H}^\flat{}_\alpha \approx -h^\flat{}_\alpha{}^{\overline{k}}\eta^{\flat\overline{\bar{j}l}}\mathcal{D}^\flat{}_{\bar{j}}\hat{\hat{\pi}}^\flat{}_{\overline{kl}} - h^\flat{}_\alpha{}^{\overline{k}}\eta^{\flat\overline{\bar{j}l}}\mathcal{D}^\flat{}_{\bar{j}}\widetilde{\hat{\pi}}^\flat{}_{\overline{kl}}, \tag{71b}$$

$$\mathcal{H}^\flat{}_{\overline{kl}} \approx 2\hat{\hat{\pi}}^\flat{}_{\overline{kl}} + \mathcal{D}^\flat{}_{[\overline{k}}\vec{\hat{\pi}}^\flat{}_{\bar{l}]}, \tag{71c}$$

$$\mathcal{H}^\flat{}_{\perp\overline{k}} \approx \hat{\pi}^\flat{}_{\perp\overline{k}} + \eta^{\flat\overline{\bar{j}l}}\mathcal{D}^\flat{}_{\bar{j}}\hat{\hat{\pi}}^\flat{}_{\perp\overline{kl}}. \tag{71d}$$

The conjugate pairs together eliminate six, two and 10 D.o.F before terminating. As with Case 28, $\hat{\pi}^\flat{}_{\overline{k}}$ becomes solenoidal due to $\chi^\flat{}_\perp$ and loses one D.o.F, while three D.o.F are lost by each of $\mathcal{H}^\flat{}_{\overline{kl}}$ and $\mathcal{H}^\flat{}_{\perp\overline{k}}$. In total, two propagating D.o.F remain as expected from Table II

$$2 = \frac{1}{2}\big(80 - 2\times 10[\text{sPFC}] - 2\times(3+3)[\text{sSFC}] - 2\times(1+1+5)[\text{iPFC}] - (3+1+5)[\text{iPSC}] - 2\times(1+5)[\text{iSFC}] - (3+1+5)[\text{iSSC}]\big). \tag{72}$$

After some further calculations, we identify the following multipliers

$$\mathcal{D}^\flat{}_{[\overline{k}|}\eta^{\flat\overline{\bar{i}\bar{j}}}\mathcal{D}^\flat{}_{\bar{i}}\big(\widetilde{\hat{\pi}}^\flat{}_{|\bar{l}]\bar{j}} + \hat{\hat{\pi}}^\flat{}_{|\bar{l}]\bar{j}}\big) + \frac{\hat{\beta}_3}{\hat{\alpha}_5}\mathcal{D}^\flat{}_{[\overline{k}}\vec{\hat{\pi}}^\flat{}_{\bar{l}]} - 2\hat{\beta}_3\eta^{\flat\overline{\bar{i}\bar{j}}}\mathcal{D}^\flat{}_{\bar{i}}\mathcal{D}^\flat{}_{\bar{j}}\mathcal{T}^\flat{}_{\perp\overline{kl}} + \hat{\beta}_3\eta^{\flat\overline{\bar{i}\bar{j}}}\mathcal{D}^\flat{}_{\bar{i}}\big(7\mathcal{R}^\flat{}_{\perp\overline{\bar{j}kl}} - 8\,{}^{\mathrm{T}}\mathcal{R}^\flat{}_{\perp\overline{\bar{i}kl}}\big) + 16\frac{\hat{\beta}_3{}^2}{\hat{\alpha}_5}\hat{u}^\flat{}_{\overline{kl}} - 8\hat{\beta}_3\mathcal{D}^\flat{}_{[\overline{k}|}\eta^{\flat\overline{\bar{i}\bar{j}}}\mathcal{D}^\flat{}_{\bar{i}}\hat{u}^\flat{}_{|\bar{l}]\bar{j}} = 0, \tag{73a}$$

$$^{\mathrm{P}}u^\flat = 0, \tag{73b}$$

(73c)

B. Case 3

It should come as no surprise that Case 3 is a relaxation of Case 17, which admits an extra D.o.F. The PPM is:

$$\begin{array}{cc|ccccc|c}
 & & & \downarrow & & & \downarrow & \\
 & & \varphi & \hat{\varphi}_{\overline{kl}} & \varphi_\perp & \widetilde{\varphi}_{\perp\overline{kl}} & {}^{\mathrm{T}}\varphi_{\overline{klo}} & \\
 & \varphi & \cdot & \hat{\pi} & \cdot & \cdot & \cdot & 1 \\
\rightarrow & \hat{\varphi}_{\overline{kl}} & \hat{\pi} & \cdot & \hat{\pi} & \hat{\pi} & \hat{\pi}! & 3 \\
 & \varphi_\perp & \cdot & \hat{\pi} & \cdot & \cdot & \cdot & 1 \\
 & \widetilde{\varphi}_{\perp\overline{kl}} & \cdot & \hat{\pi} & \cdot & \cdot & \cdot & 5 \\
\rightarrow & {}^{\mathrm{T}}\varphi_{\overline{klo}} & \cdot & \hat{\pi}! & \cdot & \cdot & \cdot & 5 \\
 & & 1 & 3 & 1 & 5 & 5 &
\end{array} \tag{74}$$

It is clear that $^{\mathrm{P}}\varphi^\flat$ is no longer primarily constrained. The only change to the remaining SiCs of Case 17 is

$$\mathcal{H}^\flat{}_{\overline{kl}} \approx 2\hat{\hat{\pi}}^\flat{}_{\overline{kl}} - \frac{1}{6}\epsilon^\flat{}_{\overline{klm}\perp}\eta^{\flat\overline{mn}}\mathcal{D}^\flat{}_{\overline{n}}\,{}^{\mathrm{P}}\hat{\pi}^\flat + \mathcal{D}^\flat{}_{[\overline{k}}\vec{\hat{\pi}}^\flat{}_{\bar{l}]}, \tag{75}$$

but $\mathcal{H}^\flat{}_\alpha$ is still satisfied. Overall, only the conjugate $^{\mathrm{P}}\varphi^\flat$ and $^{\mathrm{P}}\chi^\flat$ pair are removed, leaving three propagating D.o.F as expected from Table II

$$3 = \frac{1}{2}\big(80 - 2\times 10[\text{sPFC}] - 2\times(3+3)[\text{sSFC}] - 2\times(1+1+5)[\text{iPFC}] - (3+5)[\text{iPSC}] - 2\times(1+5)[\text{iSFC}] - (3+5)[\text{iSSC}]\big). \tag{76}$$

V. PHENOMENOLOGY

The results of Section IV cast serious doubts on the health of In general, this would be quite an arduous task, requiring a dedicated examination of all four equations of motion in the theory. However, we recently developed a mapping between the general quadratic torsion theory (12) and a torsion-free biscalar-tensor theory, which immediately reveals the cosmological background [10]. We begin with the spatially flat FRW interval

$$\mathrm{d}s^2 = \mathrm{d}t^2 - a^2\mathrm{d}\mathbf{x}^2, \tag{77}$$

where a is the scale factor, normalised to the contemporary epoch, from which we define the Hubble number $H \equiv \dot{a}/a$. We now align the unit timelike normal n^k to be perpendicular to the spatially flat slicing. Cosmological isotropy at the background level restricts only the 0^+ and 0^- torsion modes to propagate. From these modes respectively we define a pair of scalar fields

$$\phi \equiv \frac{2}{3}\mathcal{T}^{\overline{k}}{}_{\perp\overline{k}} - 2H, \quad \psi \equiv \frac{1}{6}\epsilon_{\bar{i}}{}^{\perp\overline{\bar{j}k}}\mathcal{T}^{\bar{i}}{}_{\overline{\bar{j}k}}\,. \tag{78}$$

These fields transform homogeneously and with the correct weight $\phi \mapsto \Omega^{-1}\phi$, $\psi \mapsto \Omega^{-1}\psi$ under changes of physical scale $b^i{}_\mu \mapsto \Omega b^i{}_\mu$. In the usual second-order formulation of gravity on the curved spacetime $\mathcal{M}$, it can

be shown that the theory

$$
\begin{aligned}
L_{\rm T} \simeq & \Big[\hat{\beta}_2 {m_{\rm p}}^2 + \frac{1}{4}(\hat{\alpha}_4+\hat{\alpha}_6)\phi^2 - \frac{1}{4}(\hat{\alpha}_2+\hat{\alpha}_3)\psi^2\Big]R \\
& + 3(\hat{\alpha}_4+\hat{\alpha}_6)X^{\phi\phi} - 3(\hat{\alpha}_2+\hat{\alpha}_3)X^{\psi\psi} \\
& + \frac{3}{4}(\hat{\alpha}_0+2\hat{\beta}_2){m_{\rm p}}^2\phi^2 - \frac{3}{4}(\hat{\alpha}_0+8\hat{\beta}_3){m_{\rm p}}^2\psi^2 \\
& + \frac{3}{8}(\hat{\alpha}_4+\hat{\alpha}_6)\phi^4 + \frac{3}{8}(\hat{\alpha}_4+\hat{\alpha}_6)\psi^4 \\
& - \frac{3}{4}\big((\hat{\alpha}_4+\hat{\alpha}_6)+2(\hat{\alpha}_2+\hat{\alpha}_3)\big)\phi^2\psi^2 \\
& + \sqrt{|J_\mu J^\mu|} + L_{\rm M}, \qquad (79a)\\
J_\mu \equiv & \big((\hat{\alpha}_2-\hat{\alpha}_3)-(\hat{\alpha}_4-\hat{\alpha}_6)\big)\psi^3\nabla_\mu(\phi/\psi) \\
& - (\alpha_0+2\hat{\beta}_2){m_{\rm p}}^2\nabla_\mu\phi, \qquad (79b)
\end{aligned}
$$

perfectly replicates the dynamics of the FRW background. We note that the Ricci scalar R is derived from the Riemann curvature as defined in (6), and while the biscalar-tensor theory is strictly torsion-free, the behaviour of the 0^+ and 0^- modes is accurately replicated by the scalars. The theory (79) is know as the *metrical analogue* (MA) of (2), and we have translated it here into the dimensionless couplings of (13).

We will restrict our attention to the massless theories. Following a reparameterisation to the weightless scalar $\zeta = \phi/\psi$, we find the MA of Case 3 becomes

$$
\begin{aligned}
L_{\rm T} \simeq & -3\hat{\alpha}_3 X^{\psi\psi} - \frac{1}{4}\hat{\alpha}_3\psi^2 R + 3\hat{\beta}_1 {m_{\rm p}}^2\psi^2 \\
& + \hat{\alpha}_3\psi^3\sqrt{|X^{\zeta\zeta}|} - \frac{3}{4}\hat{\alpha}_3\zeta^2\psi^4 + L_{\rm M}.
\end{aligned} \qquad (80)
$$

In this frame, we see that the MA can be partitioned into two. The first three terms in (80) describe a massive but *conformally coupled* scalar ψ. The fourth and fifth terms describe a *quadratic cuscuton* ζ, which is conformally coupled by multiplication with the appropriate powers of ψ.

The quadratic cuscuton is itself remarkable for replicating the cosmological background of the Einstein–Hilbert term [16]

$$
c_1 {m_{\rm p}}^3\sqrt{|X^{\zeta\zeta}|} - c_2 {m_{\rm p}}^4\zeta^2 \simeq \frac{3{c_1}^2}{16c_2}{m_{\rm p}}^2 R, \qquad (81)
$$

This unlikely-looking relation may be verified by substituting the ζ-equation into the $g_{\mu\nu}$-equation on the LHS of (81), and comparing with the Friedmann equations that follow from the $g_{\mu\nu}$-equation on the RHS. We find that the bizarre characteristics of the cuscuton can be taken further: when we replace the Planck mass with a dynamical scalar to obtain the conformally coupled quadratic cuscuton, we replicate the cosmological background of the same scalar, conformally coupled to gravity

$$
c_1\psi^3\sqrt{|X^{\zeta\zeta}|} - c_2\psi^4\zeta^2 \simeq \frac{9{c_1}^2}{4c_2}\Big(X^{\psi\psi} + \frac{1}{12}\psi^2 R\Big). \qquad (82)
$$

This result is very satisfying, but has fatal implications for the massless theories under consideration. By applying (82), we see that the fourth and fifth terms in (80) dynamically cancel with the first and second terms: the whole kinetic structure of the analogue theory vanishes! The same problem arises in Case 17, since the extra condition $\hat{\alpha}_3 = 0$ prevents the cancelling terms from appearing even at the level of (80). In both cases, the gravitational Lagrangian responsible for the cosmological background is a pure 0^- mass, and so the theories are not viable.

Notwithstanding the complete failure of the cases at hand, the result (82)

$$
L_{\rm T} \simeq -\frac{3}{2}\hat{\alpha}_3 X^{\psi\psi} - \frac{1}{8}\hat{\alpha}_3\psi^2 R + 3\hat{\beta}_1 {m_{\rm p}}^2\psi^2 + L_{\rm M}. \qquad (83)
$$

The interpretation of two of the unitarity inequalities in Table II is now quite straightforward from an inspection of (83). The condition $\hat{\alpha}_3 < 0$ wards off a ghost, while $\hat{\beta}_1 < 0$ prevents a tachyon. Note that $\hat{\alpha}_5 < 0$ does not obviously serve to stabilise the cosmology; we suppose that it is necessary in more general spacetimes. By switching to the new variable $\xi = \sqrt{|3\hat{\alpha}_1/2|}\psi$ we then see that Case 3 has the final analogue

$$
L_{\rm T} \simeq X^{\xi\xi} + \frac{1}{12}\xi^2 R - \frac{1}{2}{m_\xi}^2\xi^2 + L_{\rm M}, \qquad (84)
$$

i.e. matter with a conformally coupled scalar, whose mass is

$$
m_\xi \equiv 2\sqrt{\frac{|\hat{\beta}_1|}{|\hat{\alpha}_3|}}m_{\rm p}. \qquad (85)
$$

It is easy to read off from Table II that in restricting to Case 17 we must set $\hat{\alpha}_3 = 0$, in which case the cosmology becomes indeterminate. Note that in drawing these conclusions, we need make no reference to the equations of motion themselves.

VI. CONCLUSIONS

It was recently shown that among all parity-preserving theories of the form $\mathcal{R}+\mathcal{R}^2+\mathcal{T}^2$, there are 58 cases which are unitary and power-counting renormalisable when linearised in the absence of source currents [5, 6]. The linearisation was done around a presumed Minkowski vacuum, since even a cosmological constant is excluded as a source. In this work we have considered the eight cases detailed in Table II. We have inspected their Hamiltonian structure under the same conditions, but in both the linear and *nonlinear* regimes. Our principle findings may be summarised as follows;

1. The claims of [5, 6] regarding the particle spectrum (mass character and propagating degrees of freedom) and unitarity (using freedom from classical

ghosts as a proxy) of the linearised theories are all confirmed.

2. With the exception of Case 20 and Case 32, the commutators of primary constraints suggest significant changes to the dynamics when progressing from the linear to nonlinear regimes. These changes indicate the activation of ghostly spin-parity sectors and (possibly) acausal propagation of information.

ACKNOWLEDGMENTS

We are grateful to Emine Şeyma Kutluk and Wei Chen Lin for very profitable conversations, and also to Yun-Cherng Lin for valuable computational advice.

Appendix A: Ghosts, ranks and signatures

In this appendix, we attempt to elaborate on the motivation of the 'positive kinetic energy test', which was tacitly employed in the previous Hamiltonian treatment of Poincaré gauge theories [9].

Consider the free, vector U(1) theory on $\check{\mathcal{M}}$, without any coupling to gravity (and with Cartesian coordinates $\gamma_{\mu\nu} \equiv \eta_{\mu\nu}$), fixed to the Feynman gauge

$$L = -\frac{1}{4}F^{\mu\nu}F_{\mu\nu} - \frac{1}{2}(\partial_\mu A^\mu)^2, \tag{A1}$$

where we have $F_{\mu\nu} \equiv 2\partial_{[\mu}A_{\nu]}$. Up to a surface term, (A1) is of course equivalent to

$$L = -\frac{1}{2}\partial_\mu A_\nu \partial^\mu A^\nu, \tag{A2}$$

which safely propagates four massless polarisations, without developing any classical instability

$$\Box A_\mu = 0. \tag{A3}$$

Notwithstanding this reasonable behaviour, we see that the Hamiltonian of (A2) is unbounded from below

$$\mathcal{H} = -\frac{1}{2}\pi_\mu \pi^\mu + \frac{1}{2}\partial_\alpha A_\mu \partial^\alpha A^\mu, \tag{A4}$$

where the momentum is $\pi_\mu \equiv -\partial_0 A_\mu$, since the independent timelike polarisation will have a strictly *negative* contribution. This is naturally revealed in the $3+1$ picture, which we construct by defining a constant unit timelike normal $n^\mu n_\mu \equiv 1$, and (extending our previous overbar notation to holonomic indices) decomposing quantities into the 0^+ and 1^- irreps

$$A_\mu \equiv A_\perp n_\mu + A_{\overline{\mu}}, \quad \pi_\mu \equiv \pi_\perp n_\mu + \pi_{\overline{\mu}}. \tag{A5}$$

The Hamiltonian then separates into

$$\begin{aligned}\mathcal{H} = &-\frac{1}{2}{\pi_\perp}^2 + \frac{1}{2}\partial_\alpha A_\perp \partial^\alpha A_\perp \\ &- \frac{1}{2}\pi_{\overline{\mu}}\pi^{\overline{\mu}} + \frac{1}{2}\partial_\alpha A_{\overline{\mu}}\partial^\alpha A^{\overline{\mu}},\end{aligned} \tag{A6}$$

where the first and last pairs of terms are respectively negative and positive-definite on the null shell defined by (A3). The physical consequence is a loss of unitary: the timelike states have negative norm. In the U(1) theory, this is usually fixed by imposing a Gupta–Bleuler condition on the physical states, which is acceptable since the gauge-fixing term in (A1) was added by hand anyway. However, in the theories of gravity under consideration, the validity of a Gupta–Bleuler condition is not certain. We note that in the kinetic Hamiltonia of Eqs. (31), (41), (46), (51), (56), (62) and (68), we encounter mixed quadratic forms in the momenta, just as we do with the first and third terms of (A6). If such terms are negative-definite and propagating, we tentatively identify them with a loss of unitarity. We note that without full knowledge of both the nonlinear shell and the remaining field parts of the Hamiltonian (c.f. second and fourth terms in (A6)), this is quite dangerous. Moreover, as is evident from (A3), such negative-energy sectors do not necessarily correspond to classical ghosts.

We also mention that the sign of quadratic momenta in the $3+1$ formulation is robust against the choice of signature (as indeed it should be). Recall that throughout this article we have used the 'West Coast' signature $(+,-,-,-)$. The sign of each such term may then be inferred by the tensor rank of the momentum irrep, since every contraction on parallel indices introduces a factor of -1. Had we chosen the 'East Coast' signature $(-,+,+,+)$, these factors would not arise. Instead, we would have $n^\mu n_\mu \equiv -1$, whose powers would conspire in the O(3) decomposition of momenta to have the same effect up to an overall sign in the kinetic Hamiltonian. This final sign is changed by hand in the kinetic part of the Lagrangian, as is customary when changing signature.

Appendix B: Nonlinear Poisson brackets

Case 28 In Eqs. (32a) to (32d) we provide the nonlinear commutators of Case *6 26. In this appendix we list the emmergent commutators of the other seven theories

under consideration. The commutators of Case 28 read

$$\left\{\varphi_{\perp\bar{i}}, \varphi_{\perp\bar{l}}\right\} \approx \text{RHS of (32a)}, \tag{B1a}$$

$$\left\{\varphi_{\perp\bar{i}}, \varphi_{\perp}\right\} \approx -\frac{1}{J^2}\overrightarrow{\hat{\pi}}_{\bar{i}}\delta^3, \tag{B1b}$$

$$\left\{\varphi_{\perp\bar{i}}, \widetilde{\varphi}_{\perp\overline{lm}}\right\} \approx \frac{1}{2J^2}\eta_{\bar{i}\langle\bar{l}}\overrightarrow{\hat{\pi}}_{\overline{m}\rangle}\delta^3, \tag{B1c}$$

$$\left\{\varphi_{\perp\bar{i}}, {}^{\mathrm{T}}\varphi_{\overline{lmn}}\right\} \approx \frac{1}{2J^2}\Big[\eta_{\overline{in}}\hat{\hat{\pi}}_{\perp\overline{lm}} - \frac{1}{2}\eta_{\bar{i}[\bar{l}|}\hat{\hat{\pi}}_{\perp|\overline{m}]\overline{n}} - \frac{3}{4}\eta_{[\bar{l}|\overline{n}}\hat{\hat{\pi}}_{\perp\bar{i}|\overline{m}]}\Big]\delta^3, \tag{B1d}$$

$$\left\{\widetilde{\varphi}_{\overline{ij}}, \widetilde{\varphi}_{\overline{lm}}\right\} \approx \text{RHS of (32c)}, \tag{B1e}$$

$$\left\{\widetilde{\varphi}_{\overline{ij}}, \widetilde{\varphi}_{\perp\overline{lm}}\right\} \approx \text{RHS of (32c)}, \tag{B1f}$$

$$\left\{\widetilde{\varphi}_{\overline{ij}}, {}^{\mathrm{T}}\varphi_{\overline{lmn}}\right\} \approx \frac{1}{J^2}\Big[\epsilon_{\overline{ilm}\perp}\eta_{\overline{jn}}{}^{\mathrm{P}}\hat{\pi} + \eta_{\overline{ij}}\eta_{\overline{ln}}\overrightarrow{\hat{\pi}}_{\overline{m}}\Big]^{2^+\otimes 2^-}\delta^3. \tag{B1g}$$

In the RHS of (B1g), we introduce the notation on the outer parenthises to indicate some linear combination of the contents, with the required symmetry imposed on the indices: the full expression is quite long.

Case ^{*5}*25* The nonlinear commutators of Case ^{*5}25 have been encountered before

$$\left\{\widetilde{\varphi}_{\overline{ij}}, \widetilde{\varphi}_{\overline{lm}}\right\} \approx \text{RHS of (32c)}, \tag{B2a}$$

$$\left\{\widetilde{\varphi}_{\overline{ij}}, {}^{\mathrm{T}}\varphi_{\overline{lmn}}\right\} \approx \text{RHS of (32d)}. \tag{B2b}$$

Case 24 The nonlinear commutators of Case 24 have also been encountered before

$$\left\{\widetilde{\varphi}_{\overline{ij}}, \widetilde{\varphi}_{\overline{lm}}\right\} \approx \text{RHS of (32c)}, \tag{B3a}$$

$$\left\{\widetilde{\varphi}_{\overline{ij}}, \widetilde{\varphi}_{\perp\overline{lm}}\right\} \approx \frac{1}{2}\times\text{RHS of (32c)}, \tag{B3b}$$

$$\left\{\widetilde{\varphi}_{\overline{ij}}, {}^{\mathrm{T}}\varphi_{\overline{lmn}}\right\} \approx \text{RHS of (B1g)}. \tag{B3c}$$

Case 3 The nonlinear commutators of Case 3 are all new

$$\left\{\varphi, \hat{\varphi}_{\overline{lm}}\right\} \approx \frac{1}{J^2}\hat{\hat{\pi}}_{\overline{lm}}\delta^3, \tag{B4a}$$

$$\left\{\hat{\varphi}_{\overline{ij}}, \varphi_{\perp}\right\} \approx -\frac{1}{J^2}\hat{\hat{\pi}}_{\perp\overline{ij}}\delta^3, \tag{B4b}$$

$$\left\{\hat{\varphi}_{\overline{ij}}, \widetilde{\varphi}_{\perp\overline{lm}}\right\} \approx \frac{1}{2J^2}\Big[\eta_{\bar{i}\langle\bar{l}|}\hat{\hat{\pi}}_{\perp\bar{j}|\overline{m}\rangle} - \eta_{\bar{j}\langle\bar{l}|}\hat{\hat{\pi}}_{\perp\bar{i}|\overline{m}\rangle}\Big]\delta^3, \tag{B4c}$$

$$\left\{\hat{\varphi}_{\overline{ij}}, {}^{\mathrm{T}}\varphi_{\overline{lmn}}\right\} \approx \frac{1}{J^2}\Big[\epsilon_{\overline{ilm}\perp}\eta_{\overline{jn}}{}^{\mathrm{P}}\hat{\pi} + \eta_{\overline{ij}}\eta_{\overline{ln}}\overrightarrow{\hat{\pi}}_{\overline{m}}\Big]^{1^+\otimes 2^-}\delta^3. \tag{B4d}$$

Case 17 The nonlinear commutators of Case 17 are of course mostly the same as Case 3

$$\left\{\varphi, \hat{\varphi}_{\overline{lm}}\right\} \approx \text{RHS of (B4a)}, \tag{B5a}$$

$$\left\{\hat{\varphi}_{\overline{ij}}, \varphi_{\perp}\right\} \approx \text{RHS of (B4b)}, \tag{B5b}$$

$$\left\{\hat{\varphi}_{\overline{ij}}, {}^{\mathrm{P}}\varphi\right\} \approx -\frac{1}{J^2}\eta^{\overline{kl}}\epsilon_{\overline{ijk}\perp}\overrightarrow{\hat{\pi}}_{\bar{l}}\delta^3, \tag{B5c}$$

$$\left\{\hat{\varphi}_{\overline{ij}}, \widetilde{\varphi}_{\perp\overline{lm}}\right\} \approx \text{RHS of (B4c)}, \tag{B5d}$$

$$\left\{\hat{\varphi}_{\overline{ij}}, {}^{\mathrm{T}}\varphi_{\overline{lmn}}\right\} \approx \frac{1}{J^2}\Big[\eta_{\overline{ij}}\eta_{\overline{ln}}\overrightarrow{\hat{\pi}}_{\overline{m}}\Big]^{1^+\otimes 2^-}\delta^3. \tag{B5e}$$

[1] D. E. Neville, Phys. Rev. D **18**, 3535 (1978).
[2] D. E. Neville, Phys. Rev. D **21**, 867 (1980).
[3] E. Sezgin and P. van Nieuwenhuizen, Phys. Rev. D **21**, 3269 (1980).
[4] E. Sezgin, Phys. Rev. D **24**, 1677 (1981).
[5] Y.-C. Lin, M. P. Hobson, and A. N. Lasenby, Phys. Rev. D **99**, 064001 (2019), arXiv:1812.02675 [gr-qc].
[6] Y.-C. Lin, M. P. Hobson, and A. N. Lasenby, Phys. Rev. D **101**, 064038 (2020), arXiv:1910.14197 [gr-qc].
[7] M. Henneaux and C. Teitelboim, *Quantization of gauge systems* (1992).
[8] H. Chen, J. M. Nester, and H.-J. Yo, Acta Physica Polonica B **29**, 961 (1998).
[9] H.-J. Yo, J. M. Nester, and W. T. Ni, International Journal of Modern Physics D **11**, 747 (2002), arXiv:gr-qc/0112030 [gr-qc].
[10] W. E. V. Barker, A. N. Lasenby, M. P. Hobson, and W. J. Handley, Phys. Rev. D **102**, 084002 (2020), arXiv:2006.03581 [gr-qc].
[11] W. E. V. Barker, A. N. Lasenby, M. P. Hobson, and W. J. Handley, Physical Review D **102**, 024048 (2020), arXiv:2003.02690 [gr-qc].
[12] M. Blagojević, *Gravitation and Gauge Symmetries*, Series in high energy physics, cosmology, and gravitation (Institute of Physics Pub., 2002).
[13] A. N. Lasenby and M. P. Hobson, Journal of Mathematical Physics **57**, 092505 (2016), arXiv:1510.06699 [gr-qc].
[14] E. Ş. Kutluk, *Adiabatic Solutions in General Relativity and Boundary Symmetries*, Ph.D. thesis, Bogazici Uni-

versity (2019).
[15] H.-J. Yo and J. M. Nester, International Journal of Modern Physics D **8**, 459 (1999), arXiv:gr-qc/9902032 [gr-qc].
[16] N. Afshordi, D. J. H. Chung, M. Doran, and G. Geshnizjani, Phys. Rev. D **75**, 123509 (2007), arXiv:astro-ph/0702002 [astro-ph].